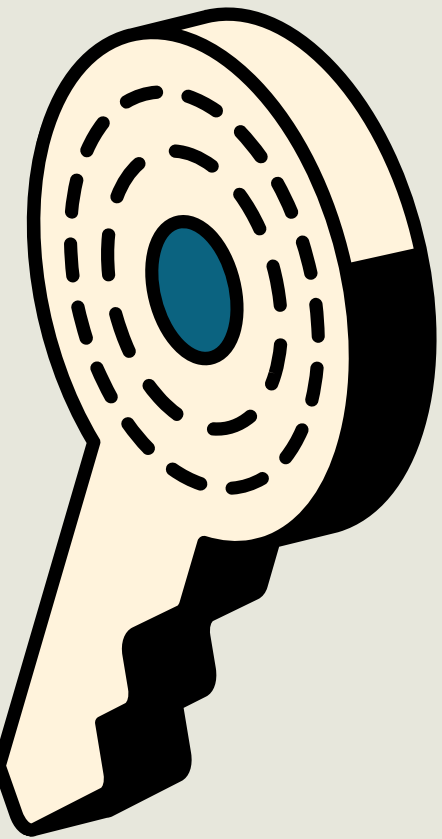
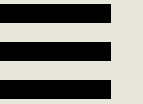
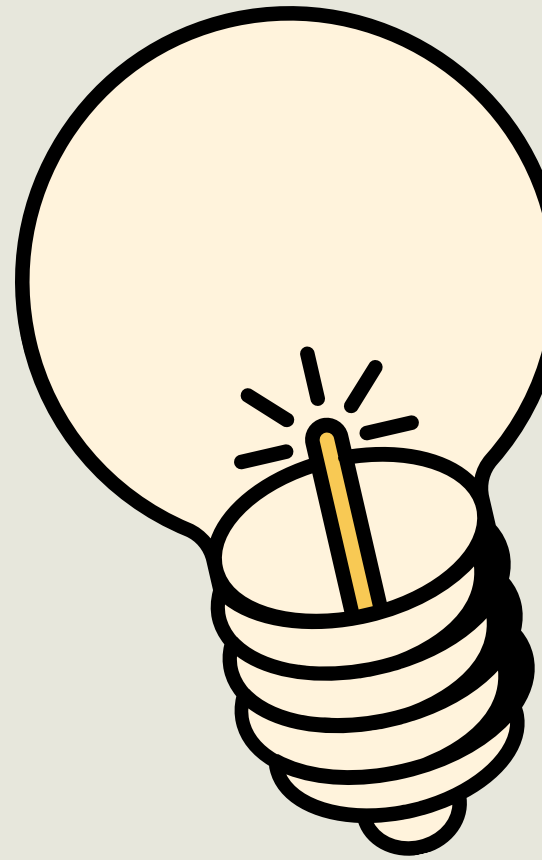


REGISTRATION



Modern WEB Development: Programming Stacks, Roles and Tools



OPENING PRAYER

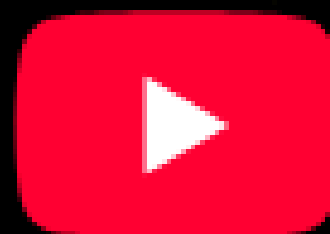


Our Lady of Fatima lyrics

Opps Paw (Opaw)



We come on bended knee



Watch on  YouTube



Lupang Hinirang - Pambansang Awit ng Pilipinas | Philippine National Anthem

Boss Kaloi

LUPANG  HINIRANG



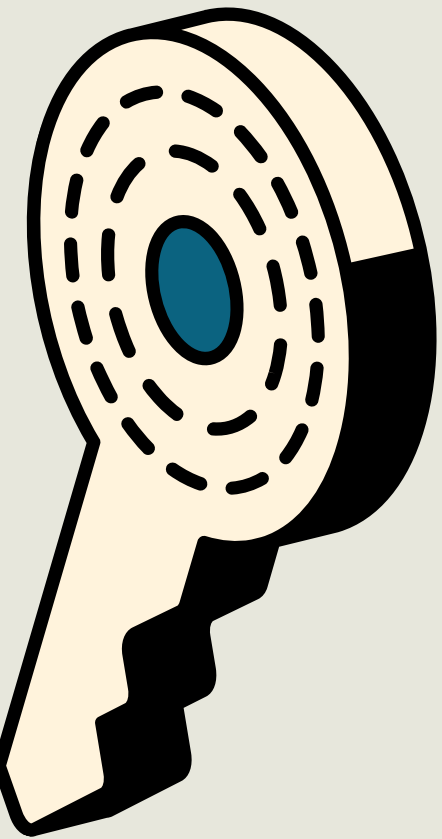
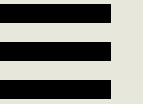
Watch on  YouTube



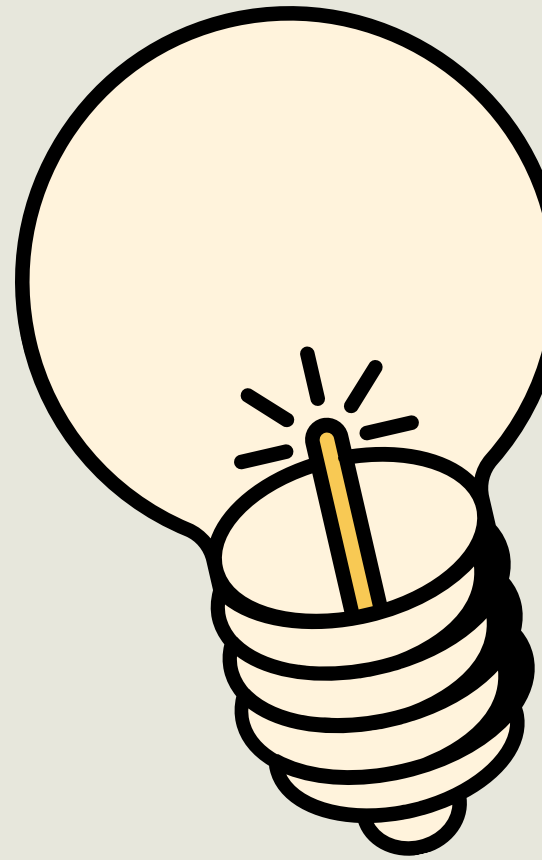
Reminders:

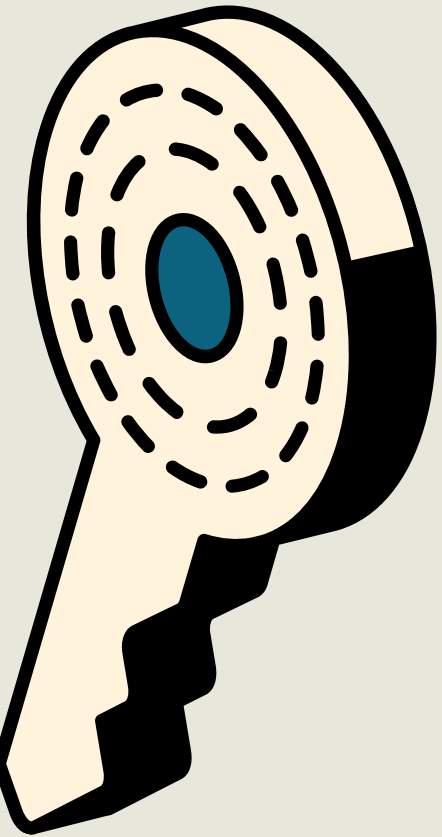
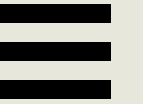


1. Silent all devices
2. Q/A will be answered after the seminar
3. Pay attention to the speaker
4. Minimize voice when talking
5. We encourage taking notes for this topic
6. No taking of photos and videos



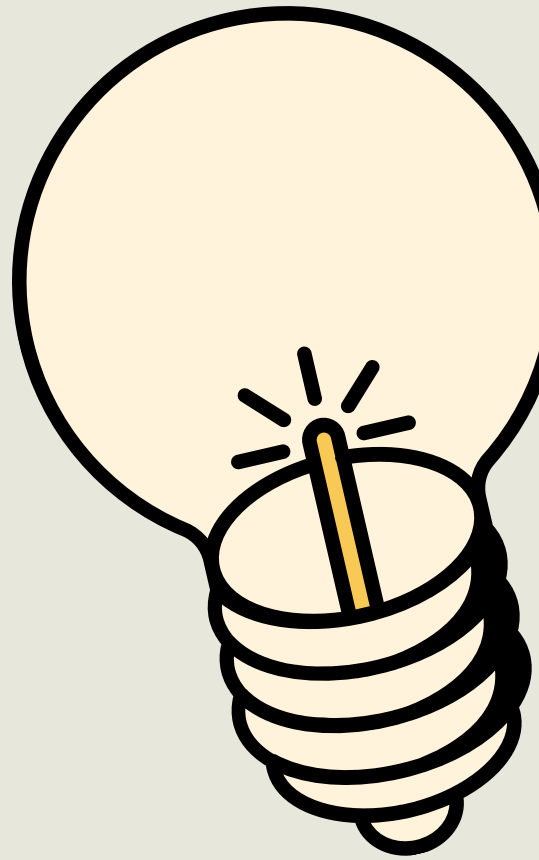
Modern WEB Development: Programming Stacks, Roles and Tools

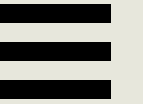




Khenard Vince L. Albaytar

**FullStack developer
(backend specialized)
Ai generation
DevOps**



The logo for Suretrack, featuring a red car icon with a white checkmark inside it.

Suretrack.

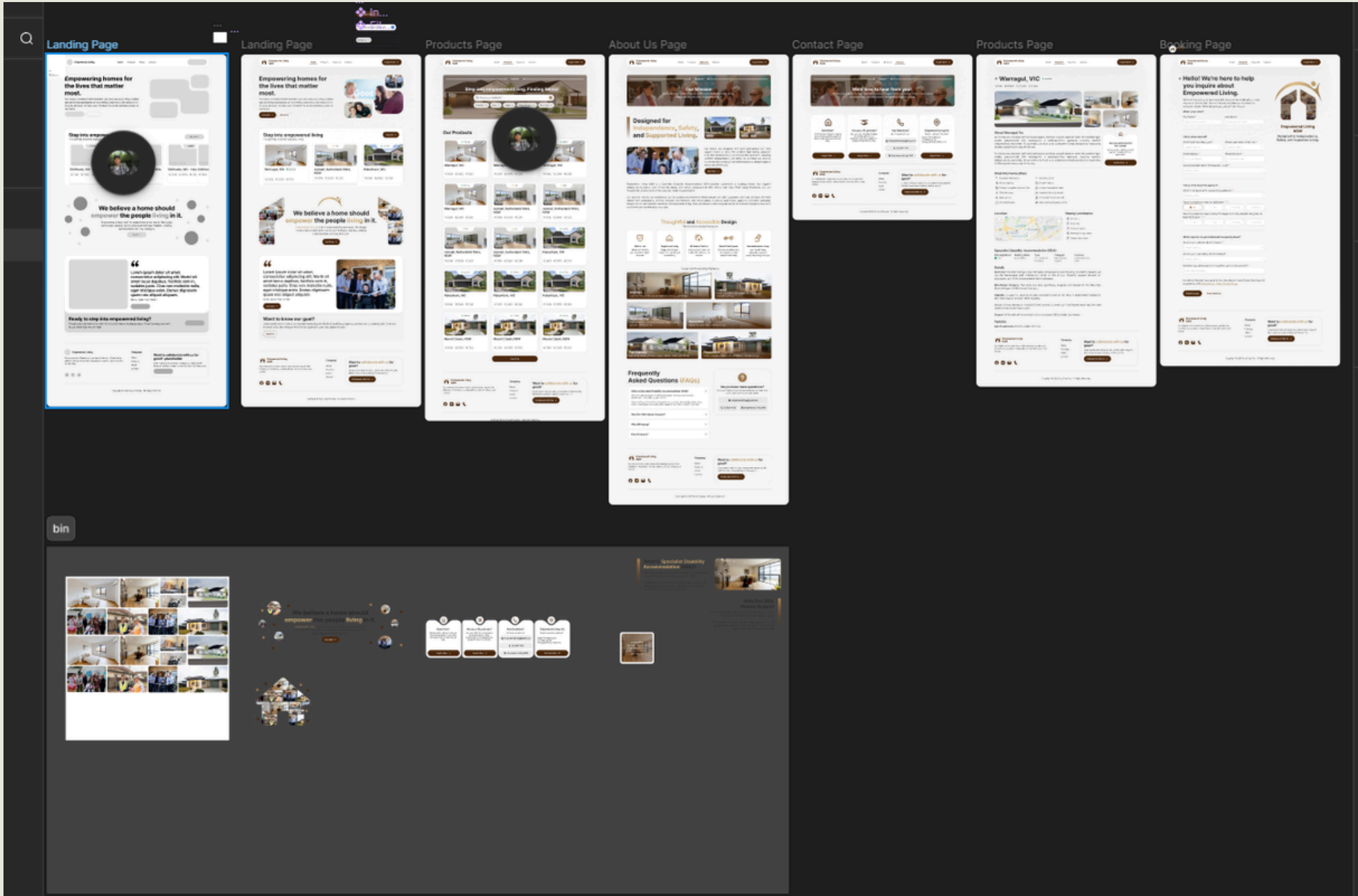
Making sure that every part, is on the track!



J2911

Accounting
Services Co.

"Your Asset in the World of Liabilities"









akotosibuboy
Khenard-Alt

Edit profile

2 followers · 2 following

Philippines

khenard.albaytar.73

Achievements



Highlights

☆ PRO

Popular repositories

Karyo Public

capstone_beatTheHeat Public

TypeScript

Student-dashboard Public

JavaScript

Customize your pins

103 contributions in the last year

Contribution settings

2026

2025

	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug
Mon												
Wed												
Fri												

Learn how we count contributions

Less More

Contribution activity

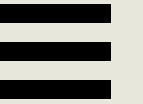
September 2026

Created 15 commits in 1 repository

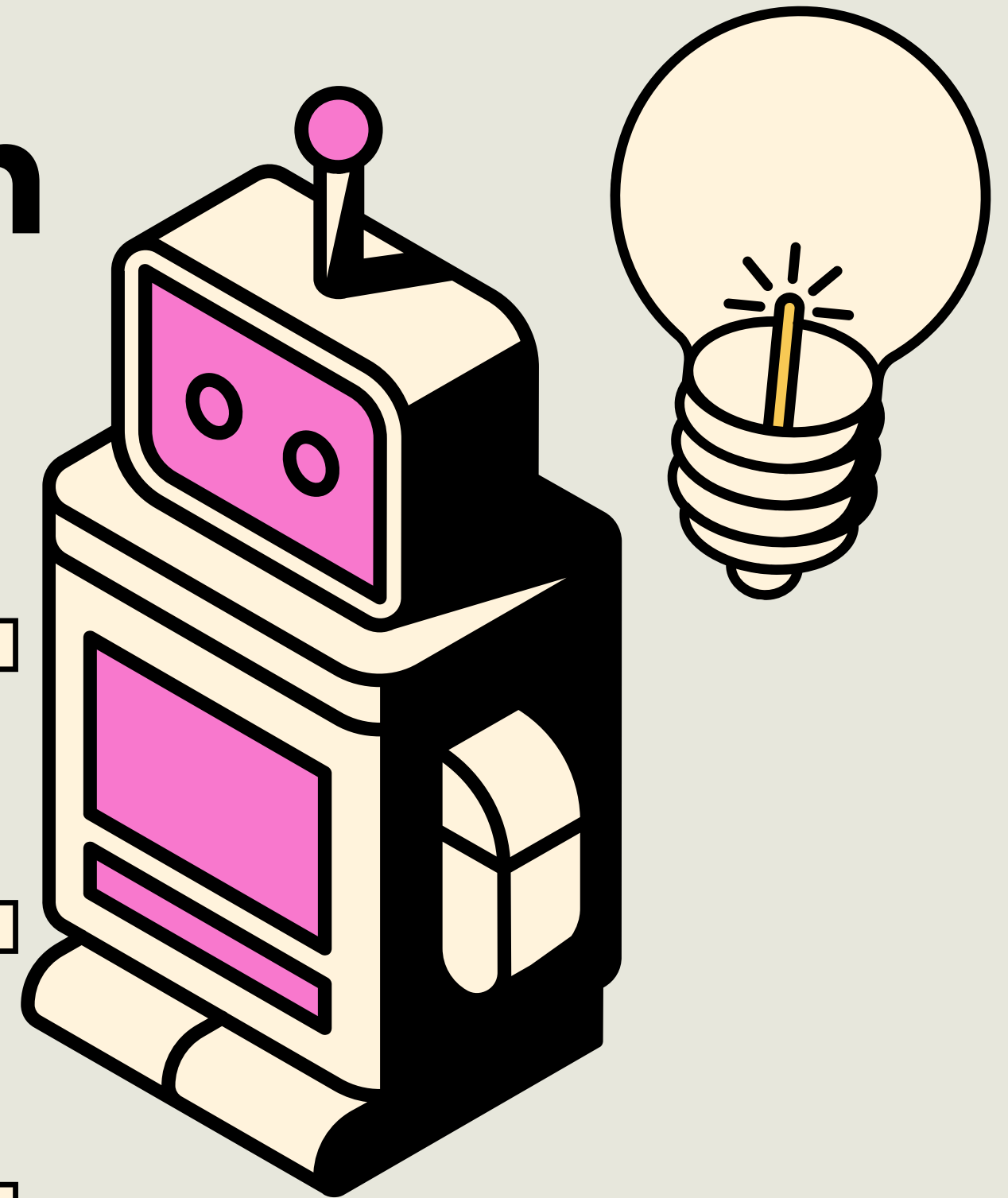
[Khenard-Alt/capstone_beatTheHeat](#) 15 commits

Show more activity

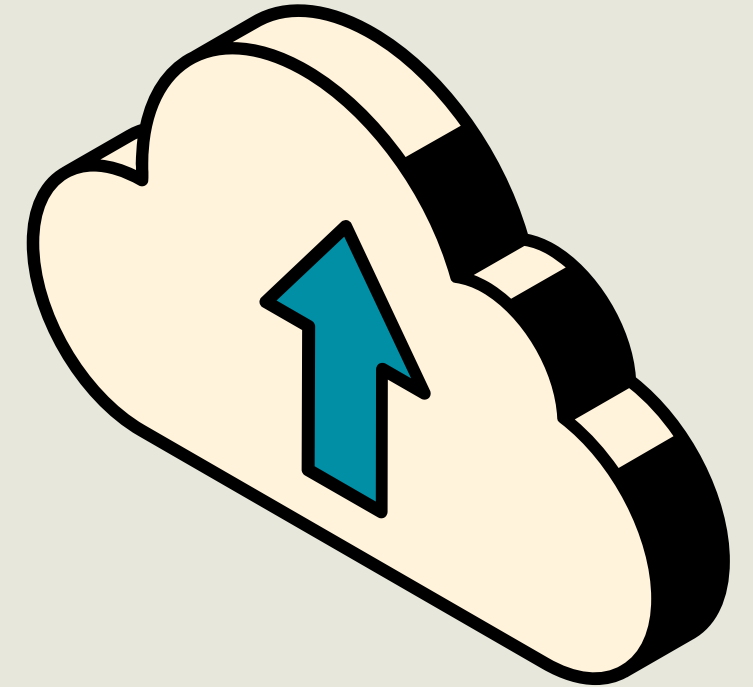
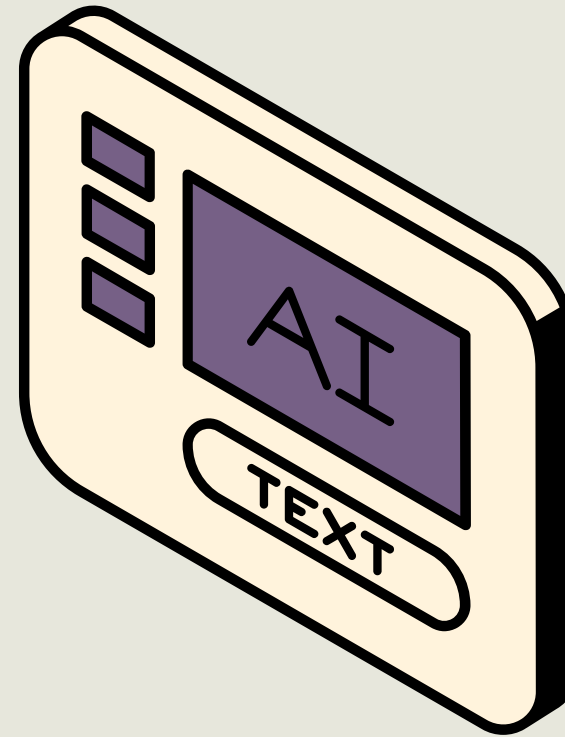
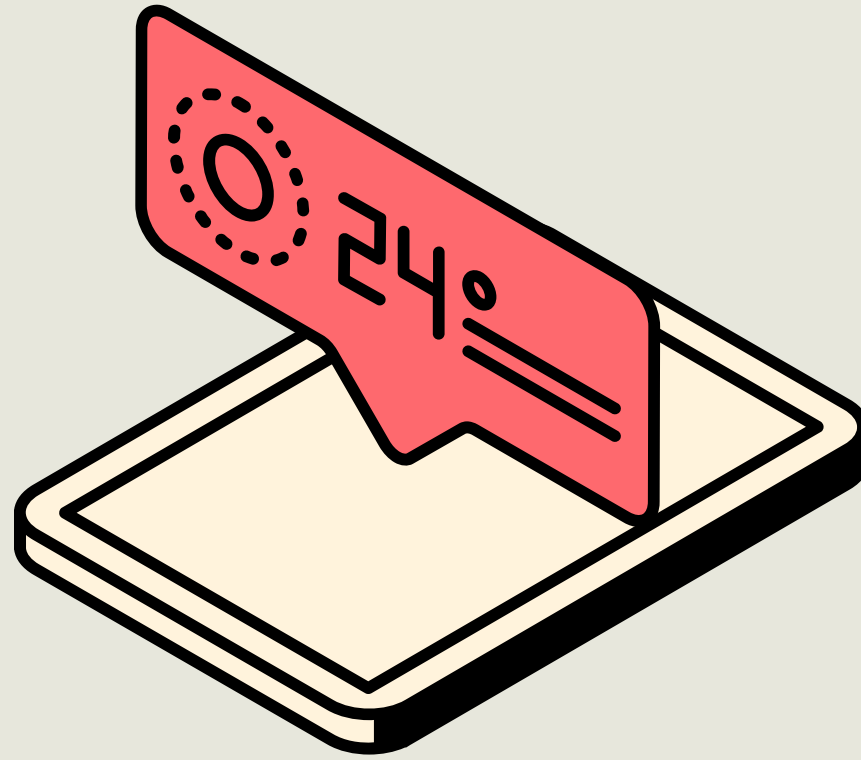
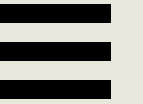
Seeing something unexpected? Take a look at the [GitHub profile guide](#).



Introduction to Modern Software Development

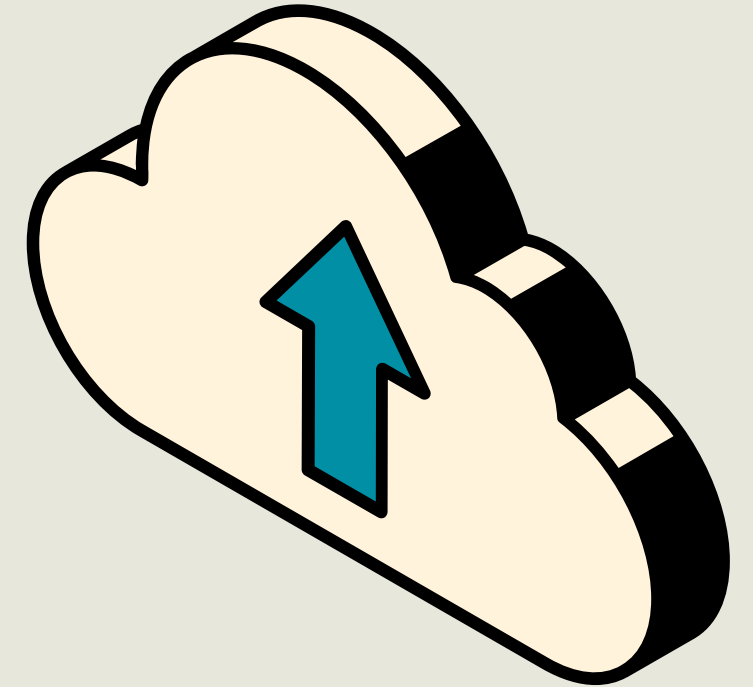
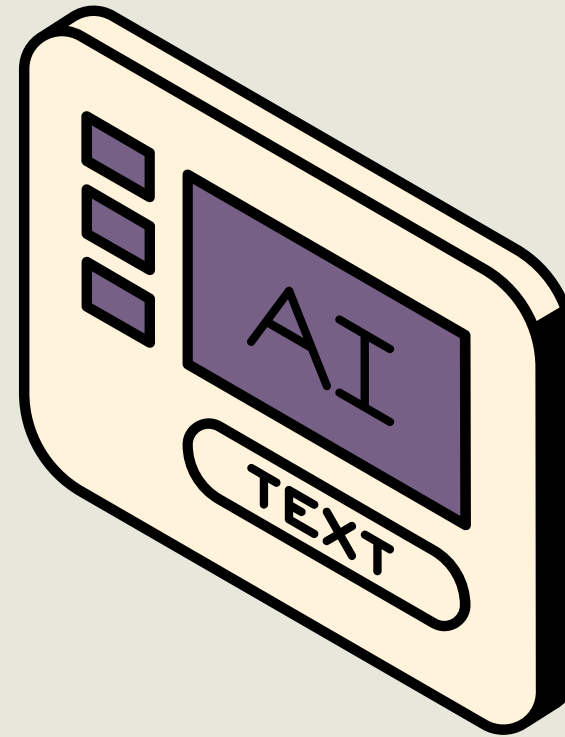
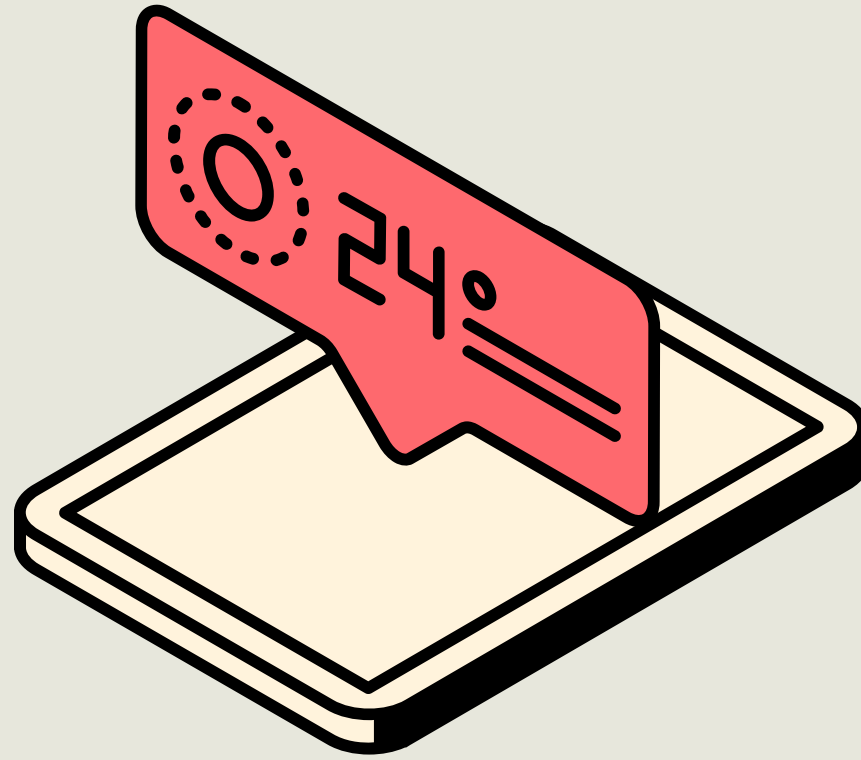
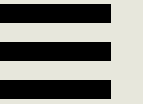


The Paradigm Shift: Software engineering has evolved from rigid, monolithic systems into cloud-native, highly modular, and AI-accelerated ecosystems.



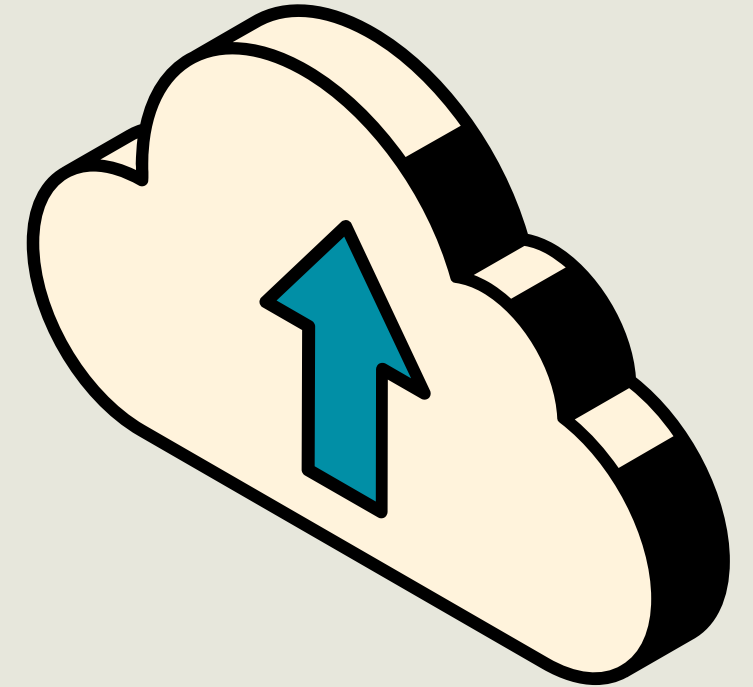
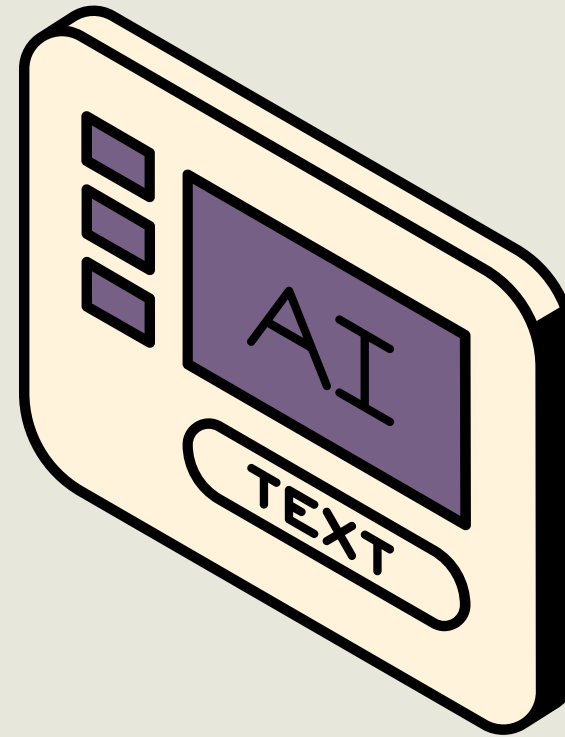
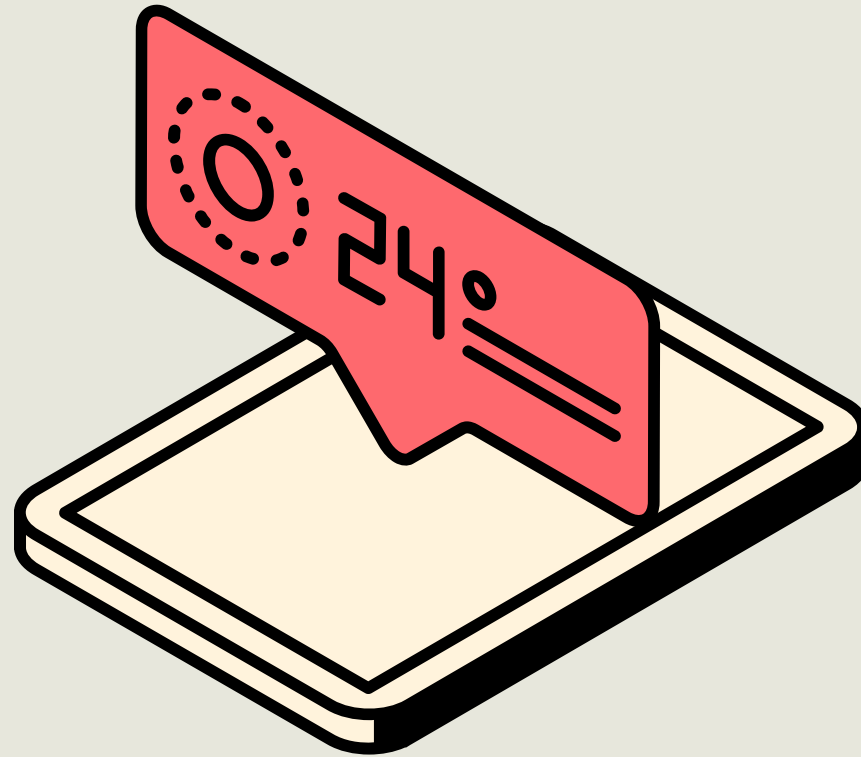
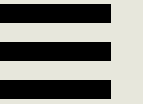
Core Drivers

Speed-to-Market: Delivering value iteratively through continuous deployment.



Core Drivers

Developer Velocity: Utilizing automation and AI to eliminate repetitive boilerplate coding.

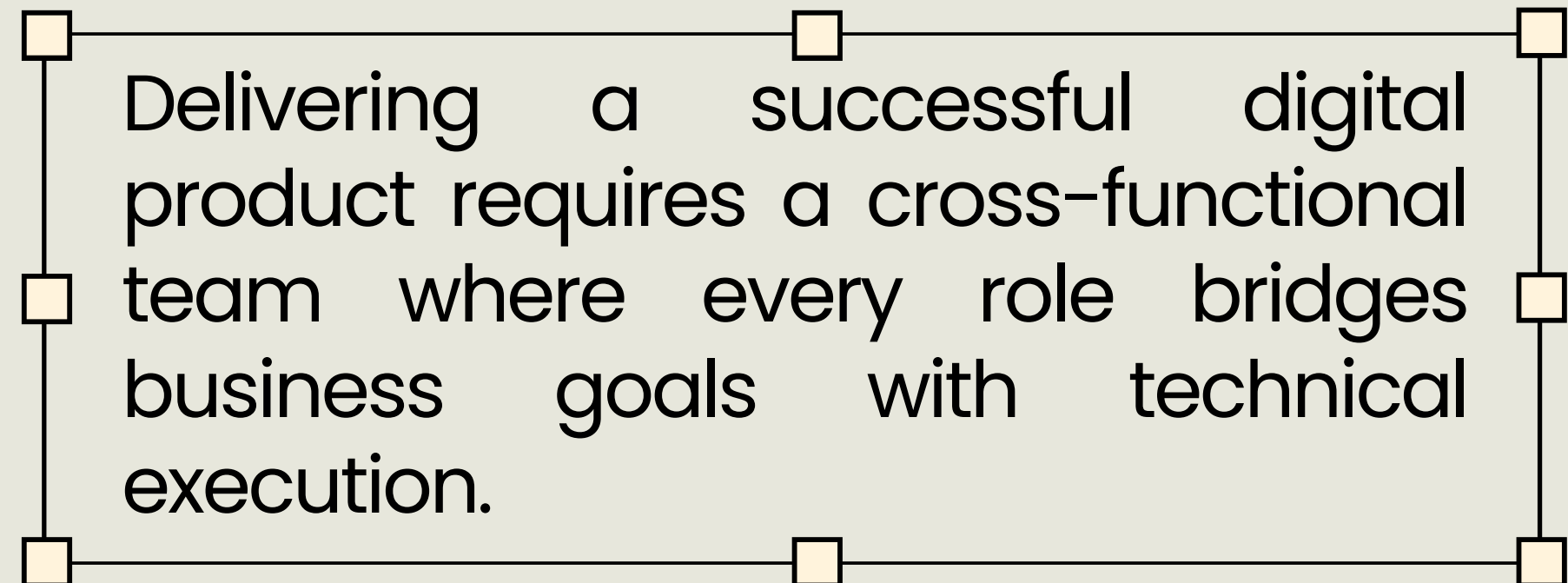


Core Drivers

The New Engineering Mindset: Modern developers are no longer just syntax writers; they are system architects, problem solvers, and integration managers who orchestrate complex tools to build reliable products.



The Modern Product Lifecycle & Team Roles

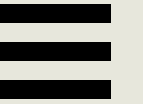


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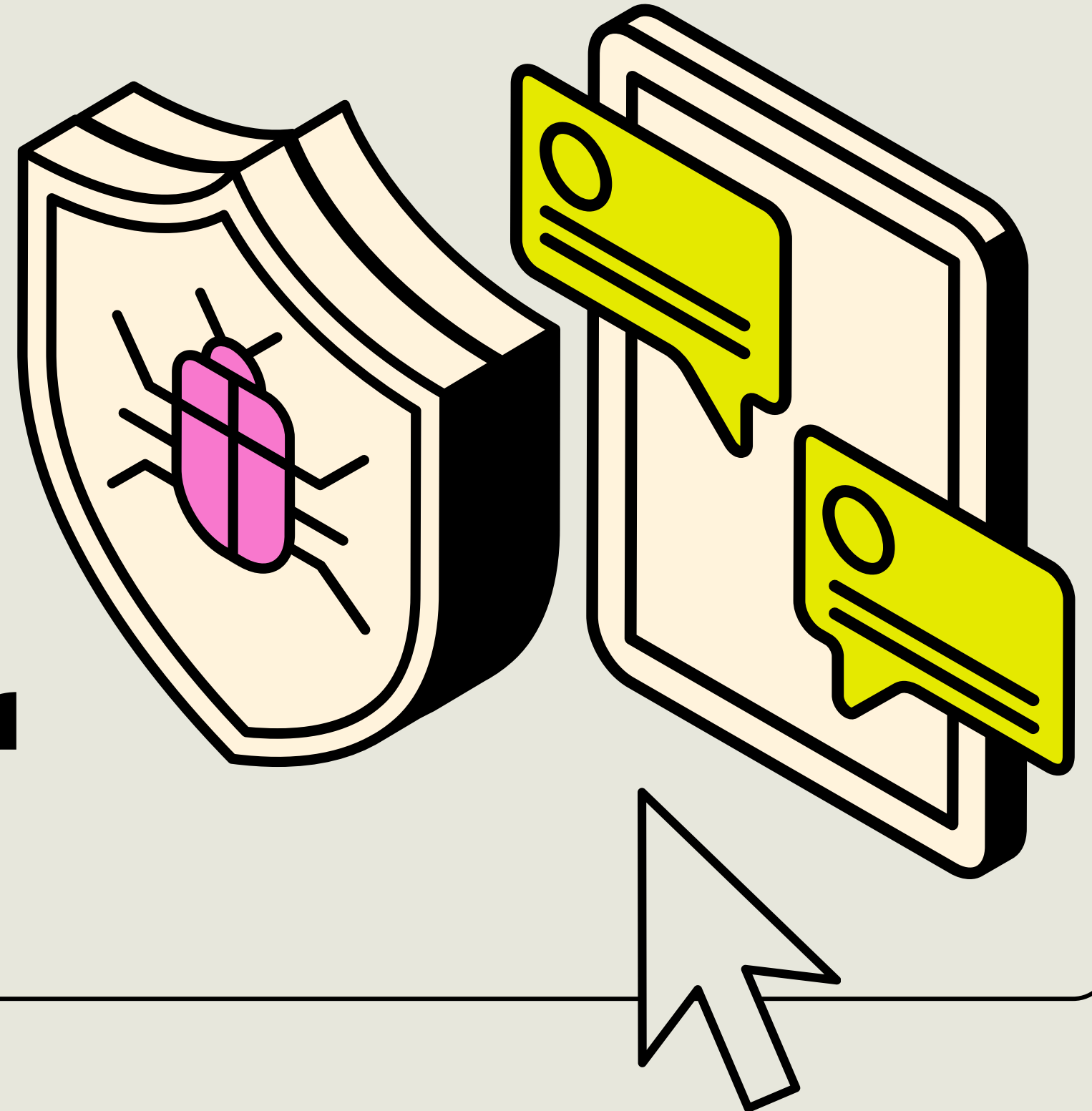
ABOUT

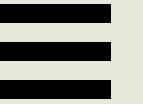
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OTHERS

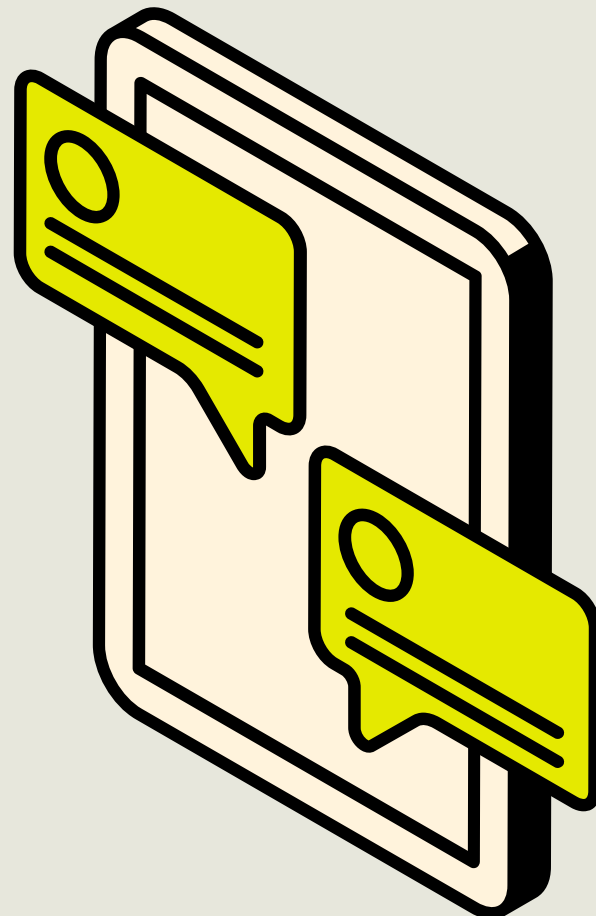
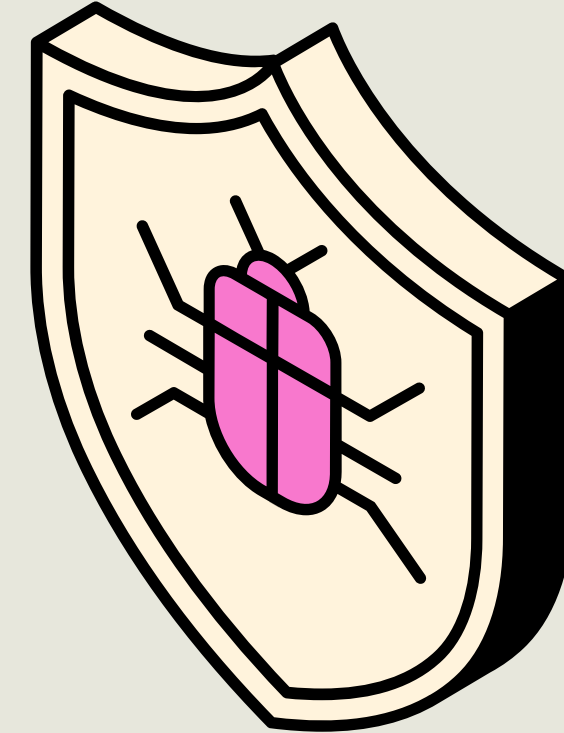


Product/ Project Manager





product vision, manages roadmaps, prioritizes backlogs, and acts as the bridge between stakeholders and engineering.



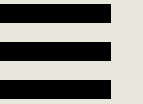
Product Requirement Documents (PRDs), User Stories, Release Roadmaps.

HOME

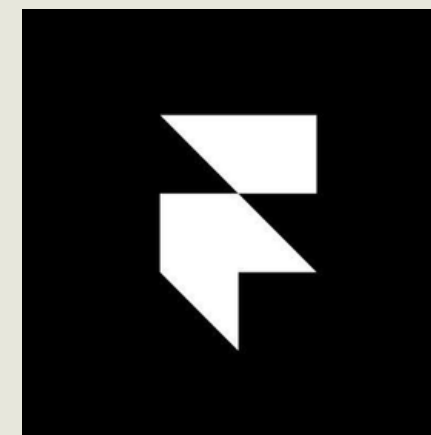
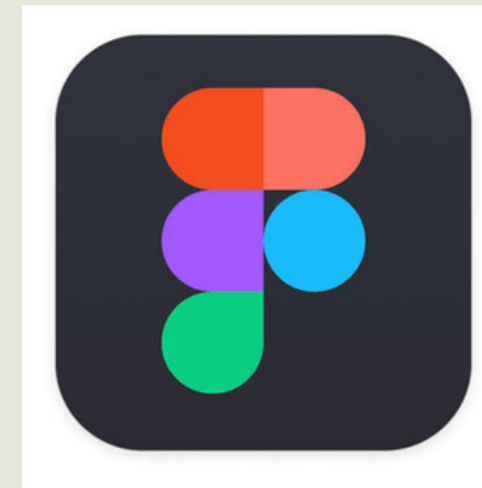
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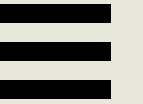
CONTENT

OTHERS



USER INTERFACE & USER EXPERIENCE





UX (User research, personas, user journeys, wireframes, usability testing) & UI (Visual hierarchy, design systems, micro-interactions, WCAG accessibility).



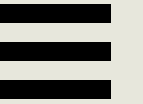
**User Flows, Wireframes, Figma
Prototypes, Design System
Tokens.**

HOME

ABOUT

CONTENT

OTHERS



FRONTEND DEVELOPER

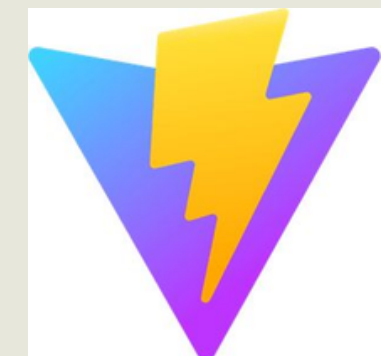
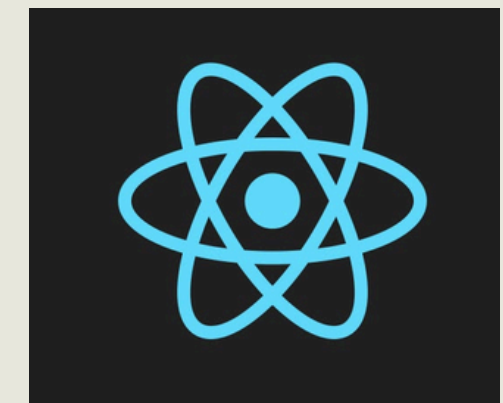
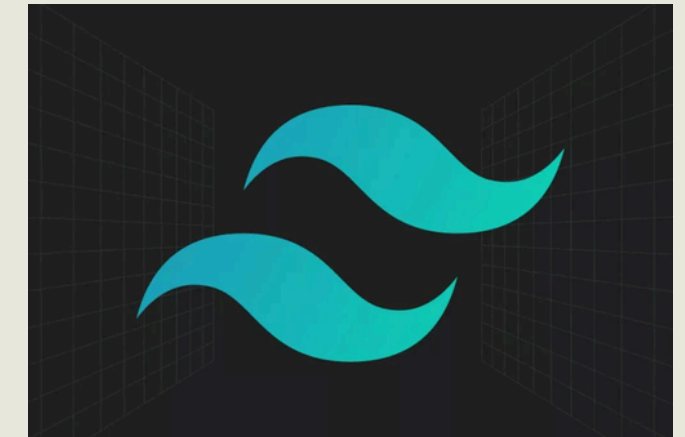
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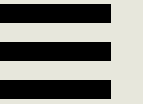


CSS

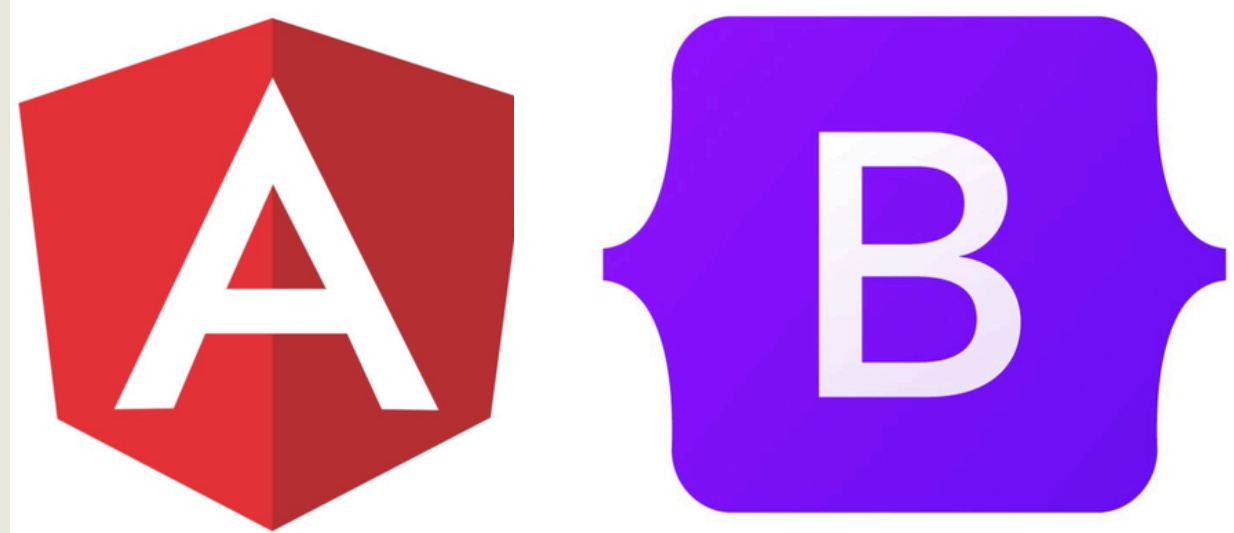


JS





Translates UI/UX designs into responsive, accessible, and high-performance client-side code.



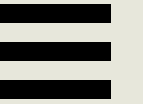
Web/Mobile Interfaces, Component Libraries, State Management.

HOME

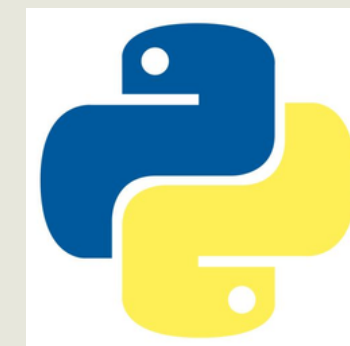
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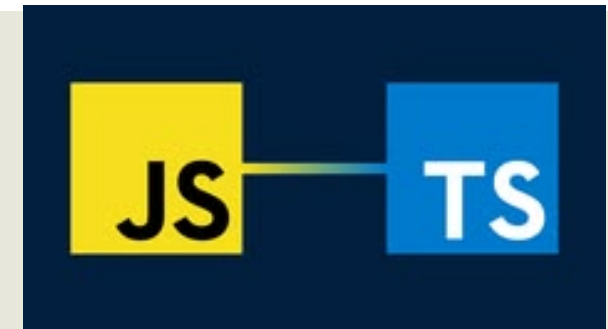
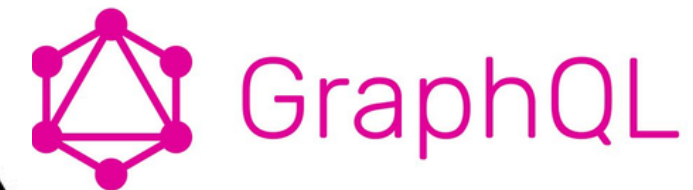


BACKEND DEVELOPER





Builds scalable server-side logic, database structures, business rules, and secure API endpoints.



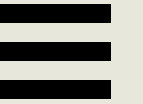
**REST/GraphQL APIs,
Database Schemas,
Authentication Systems.**

HOME

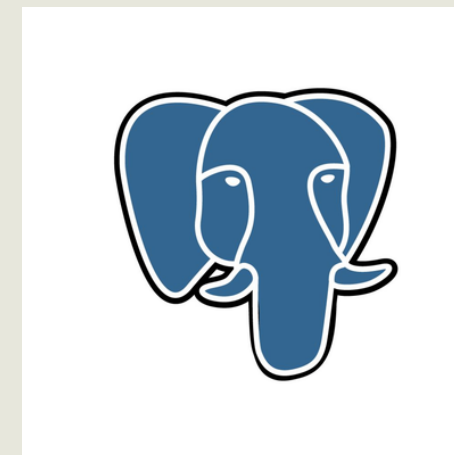
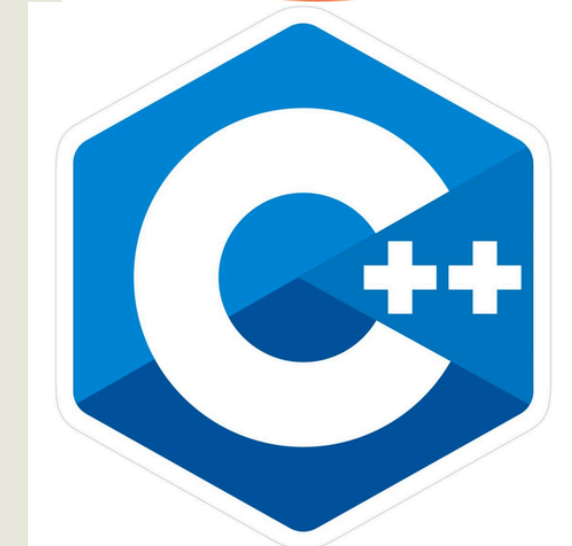
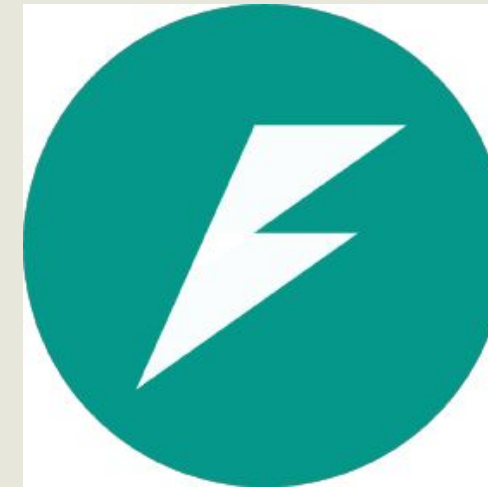
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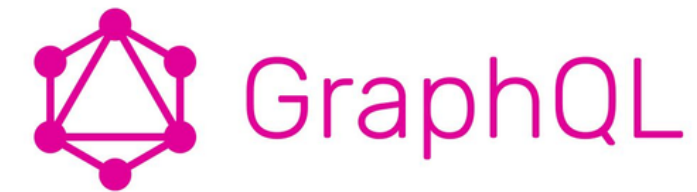


FULLSTACK DEVELOPER

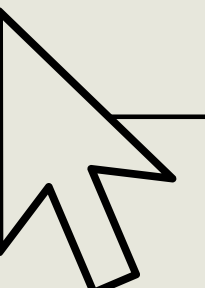




Bridges frontend and backend ecosystems; capable of building and shipping end-to-end features independently.



Full-Featured Applications, Cross-Layer Bug Fixes.

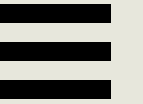


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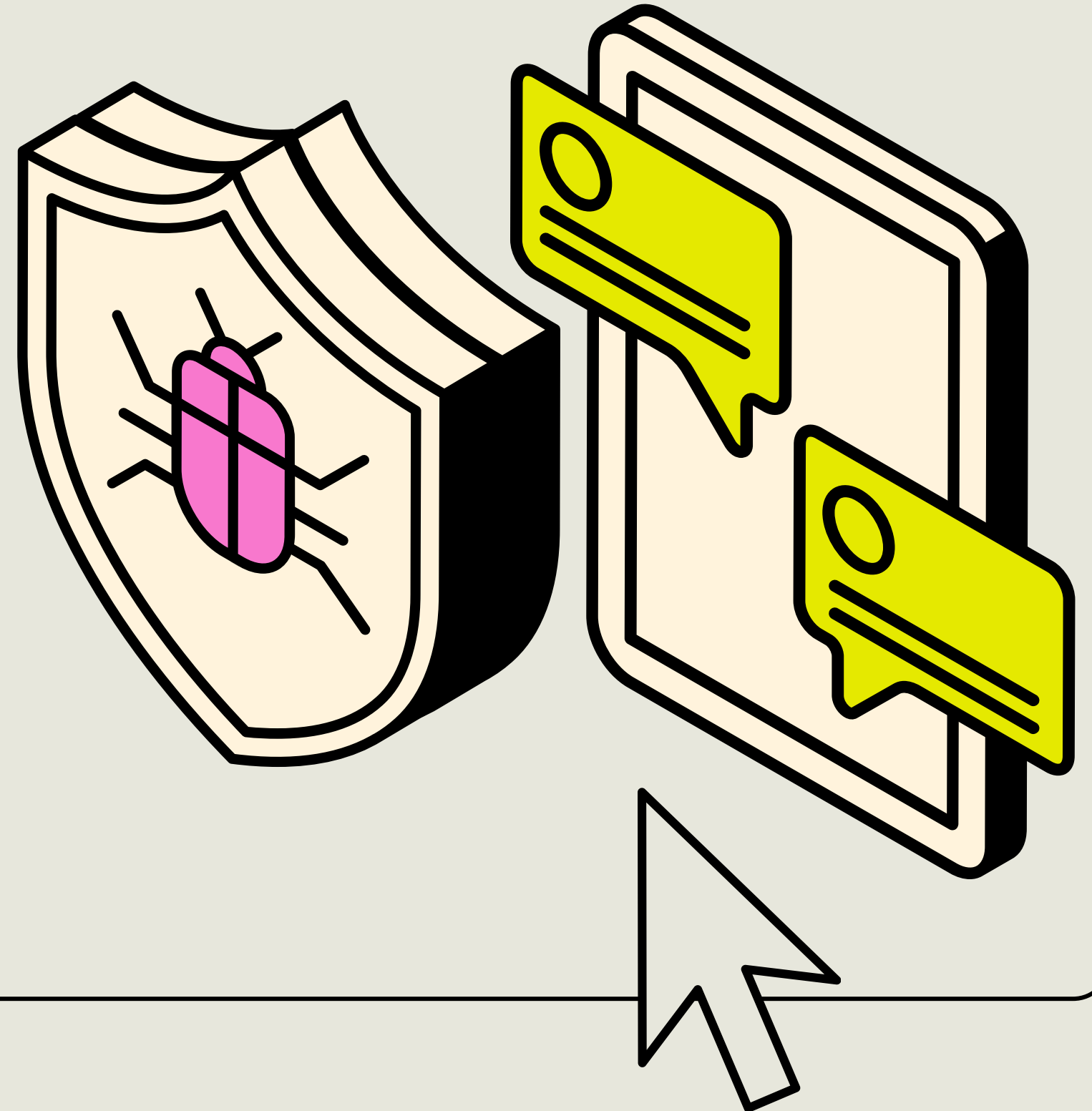
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CONTENT

OTHERS



ICE BREAKER

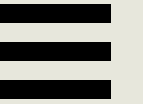


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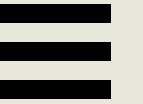
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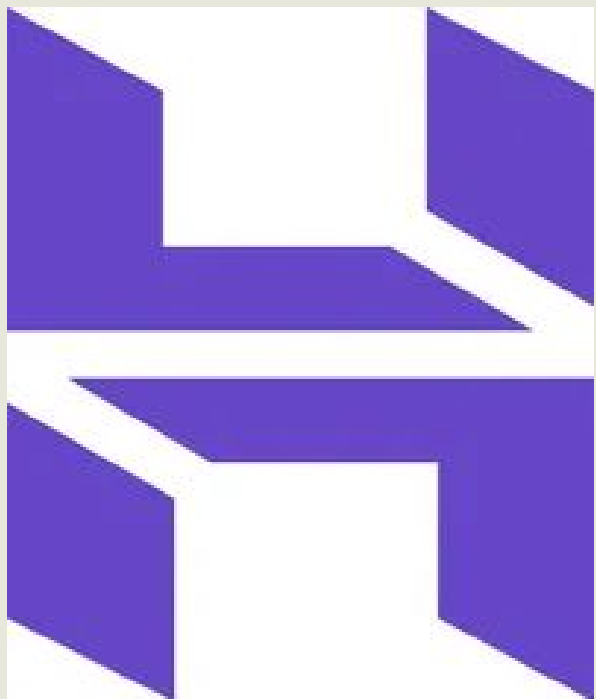
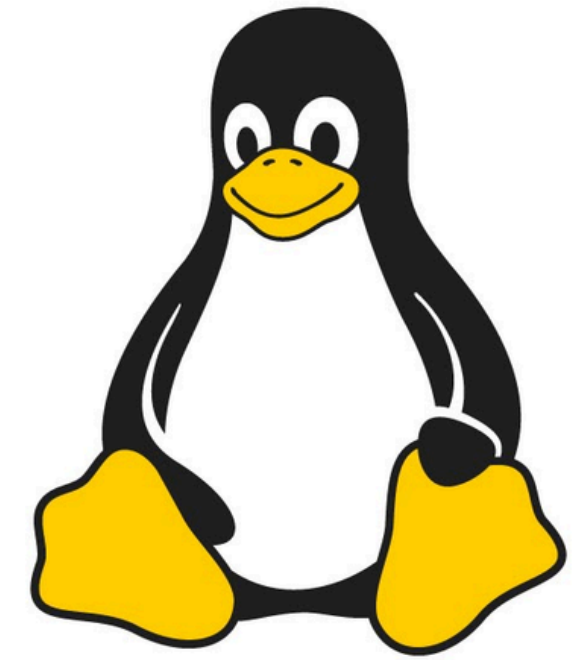
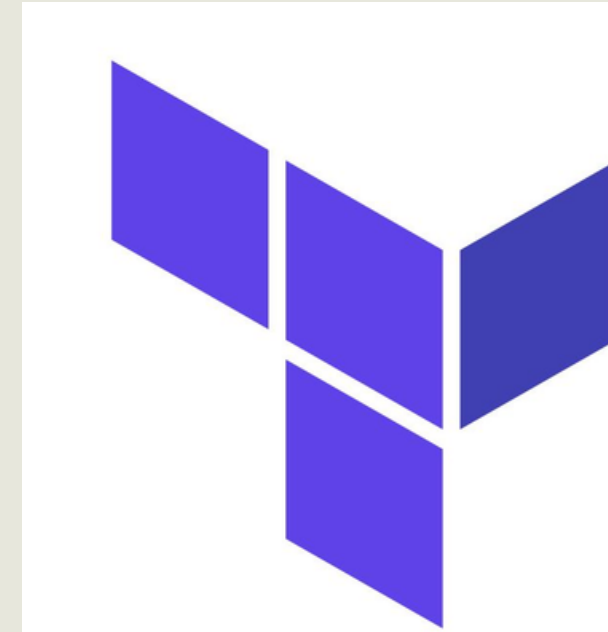
Google Cloud Platform



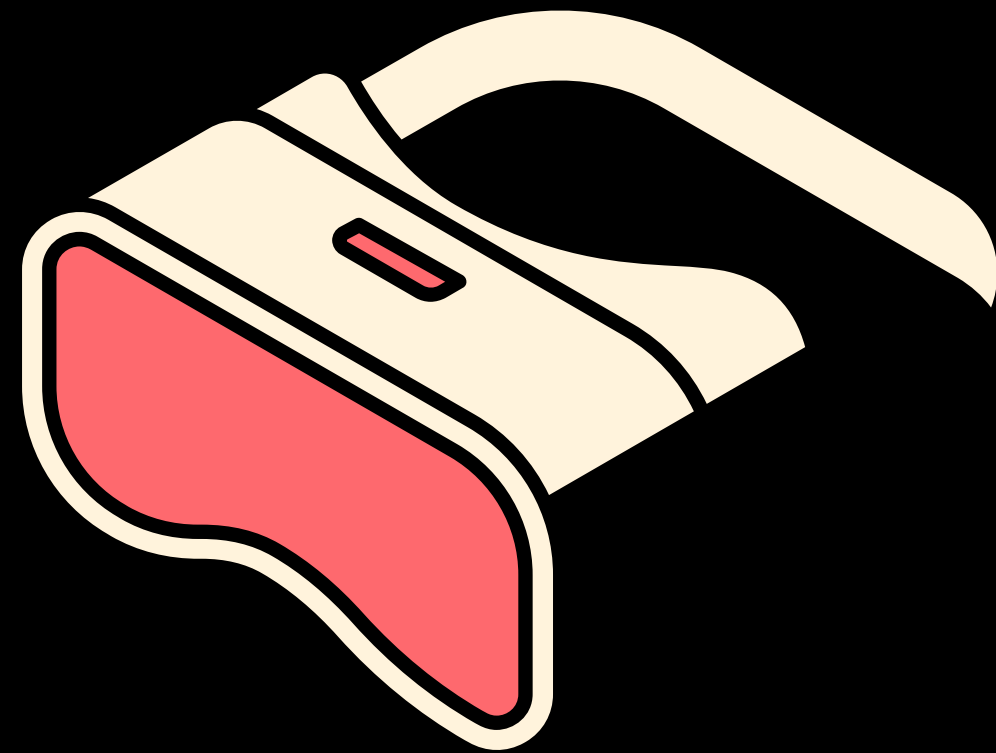
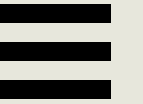
DevOps / Platform Engineer



**Automates infrastructure,
manages CI/CD pipelines,
ensures system security,
uptime, and scalability.**



**Cloud Architecture,
Automated Deployment
Pipelines, Monitoring
Dashboards.**



Anatomy of a Modern Programming Stack

A breakdown of the multi-layered technology stack driving modern web and mobile applications.

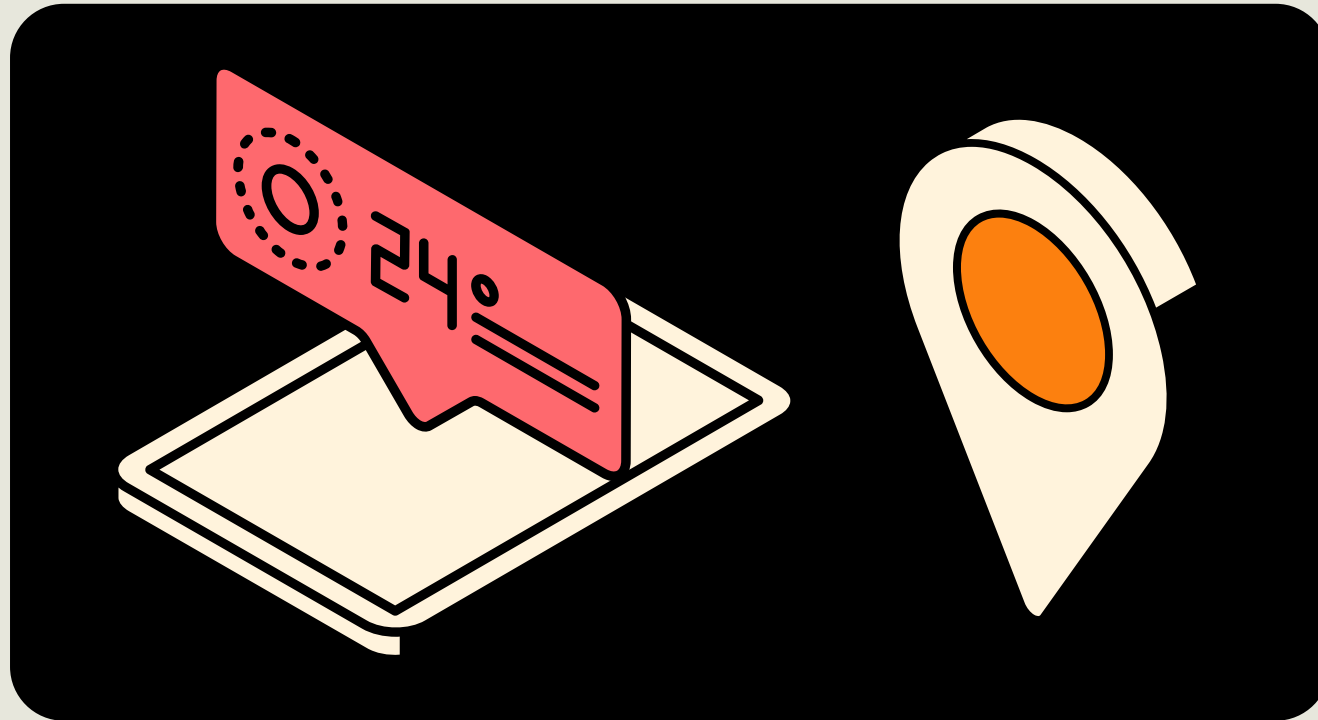
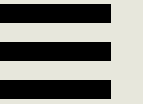


HOME

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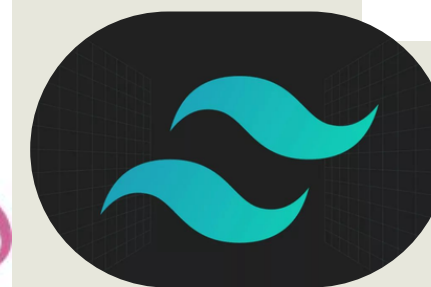


*FRONTEND LAYER
(Client-Side)*

FRAMEWORKS



STYLING



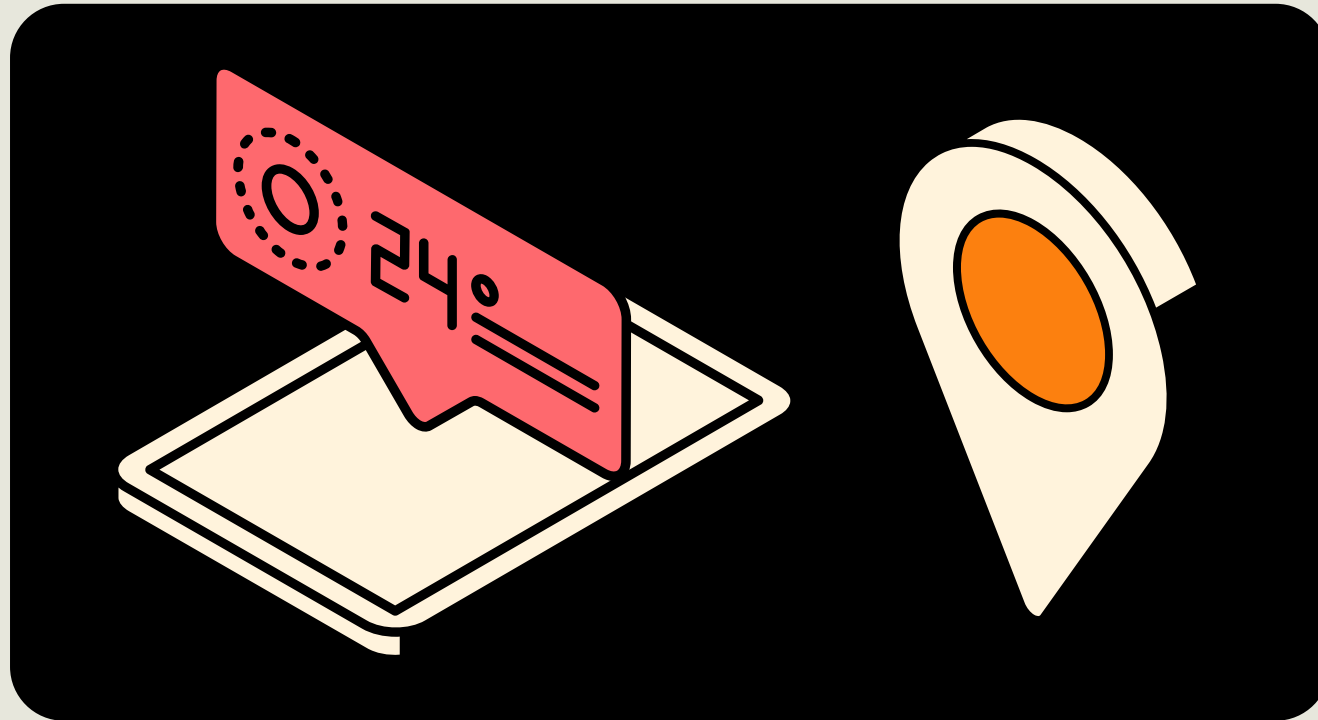
**CSS
MODULES**

HOME

ABOUT

CONTENT

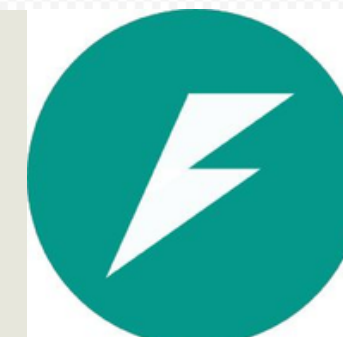
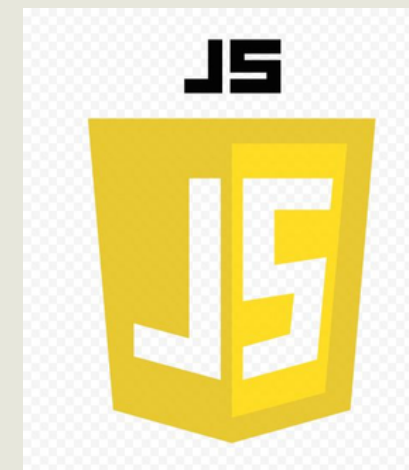
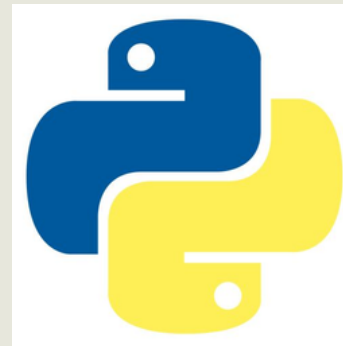
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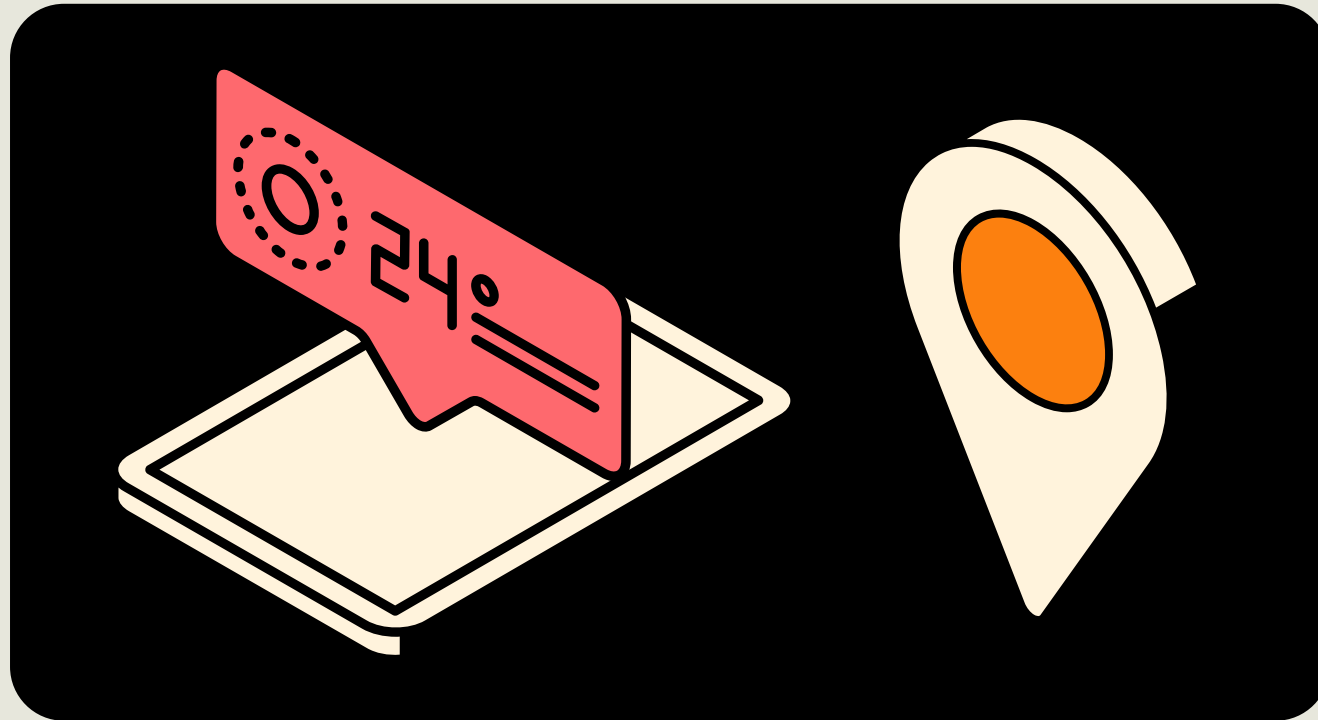
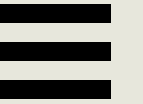


*BACKEND LAYER
(Client-Side)*

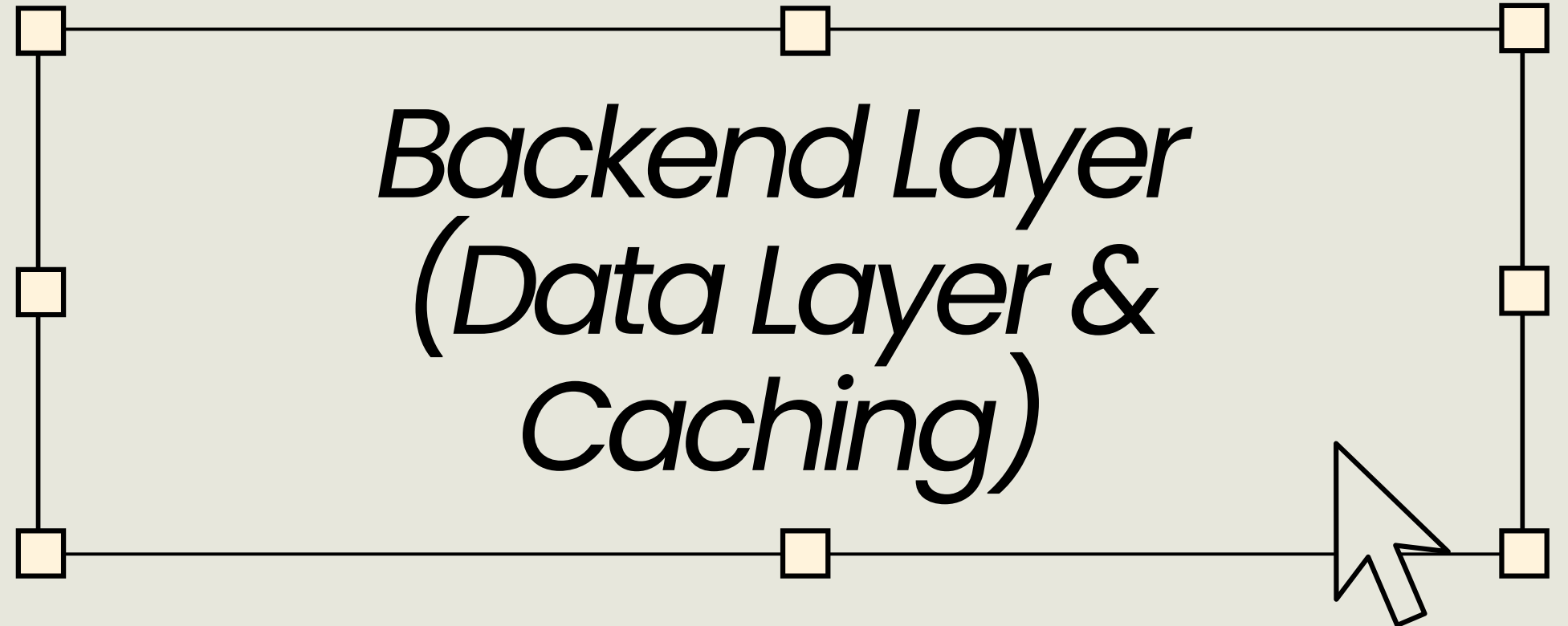
LANGUAGES

FRAMEWORKS



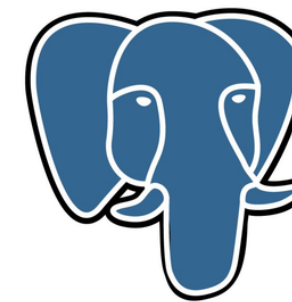


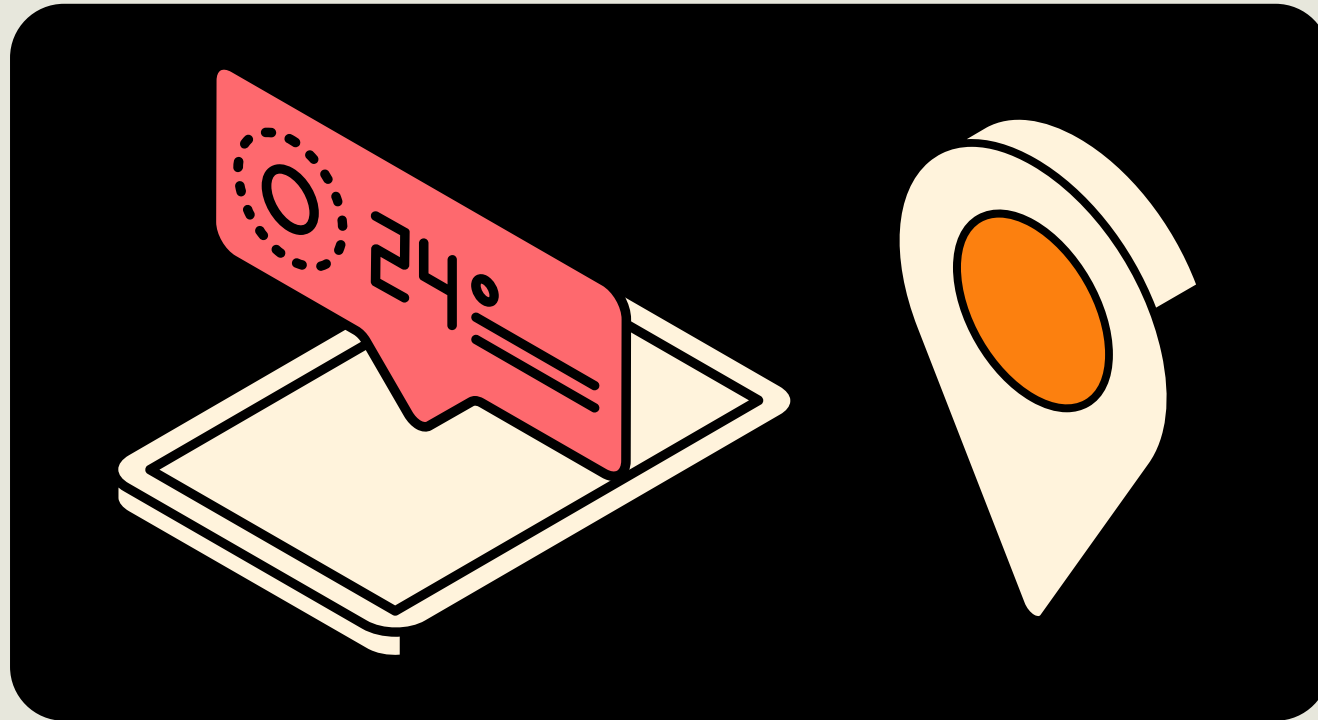
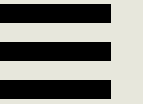
*Backend Layer
(Data Layer & Caching)*



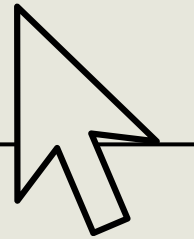
**Relational
Databases**

**No SQL
Database**





*Backend Layer
(CACHING &
IN-MEMORY)*



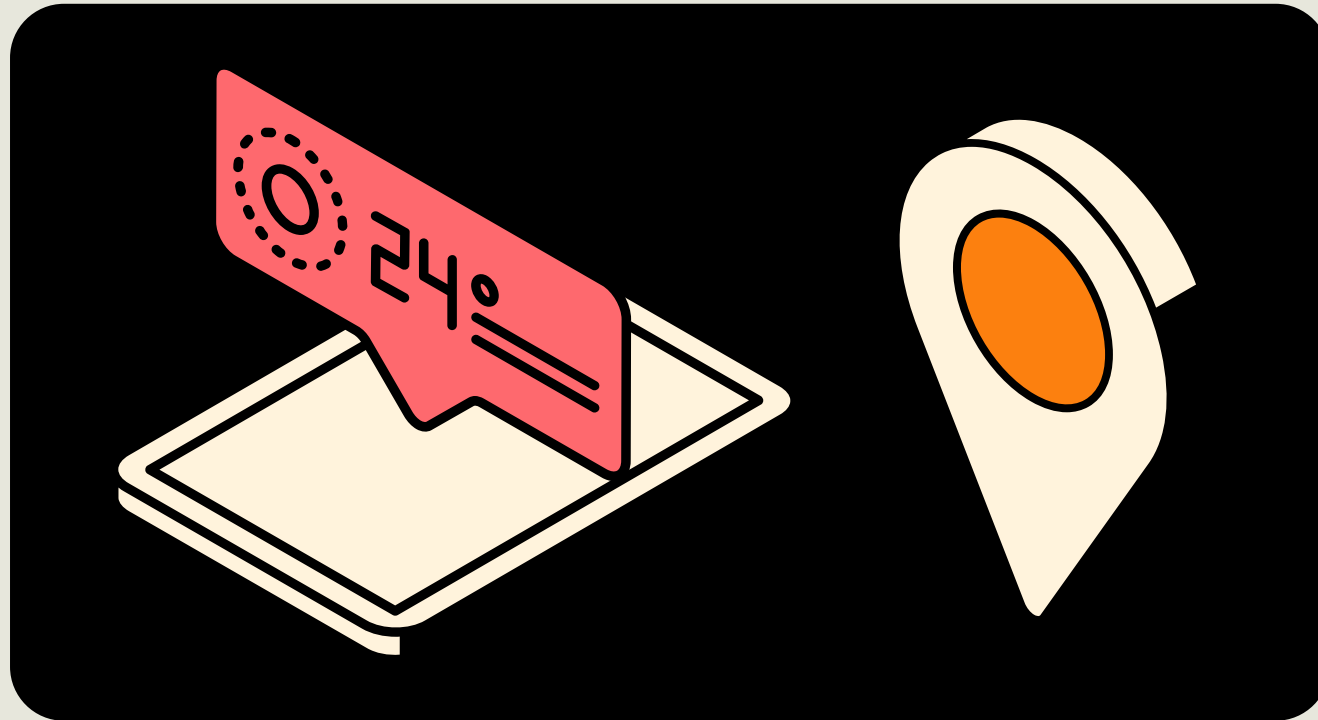
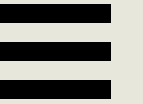
*REDIS & MEMCACHED
high-speed data retrieval
for sessions and tokens*

HOME

ABOUT

CONTENT

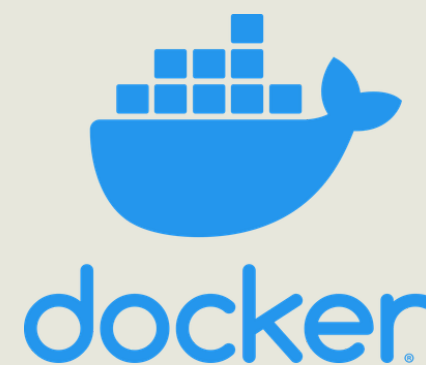
OTHERS



CLOUD AND INFRASTRUCTURE LAYER

**CLOUD
PROVIDERS**

CONTAINERIZATION



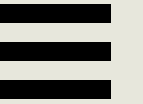


UI/UX Design and the Developer Handoff

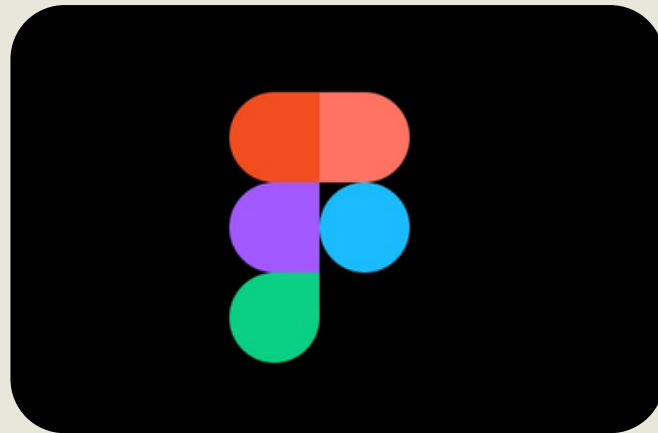


Great design means nothing if it cannot be built efficiently. The collaboration between designers and developers dictates product success.





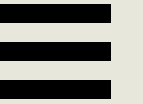
Essential Design Tools



Figma: The industry-standard tool for collaborative wireframing, UI design, and prototyping.



Design Systems: Storybook and Tailwind UI tokens that map design properties directly to code variables.



The Seamless Handoff Workflow



Research & Ideation: UX designers map out user journeys and validate ideas through wireframes.

High-Fidelity Design: UI designers apply design systems, typography, and color tokens in Figma.

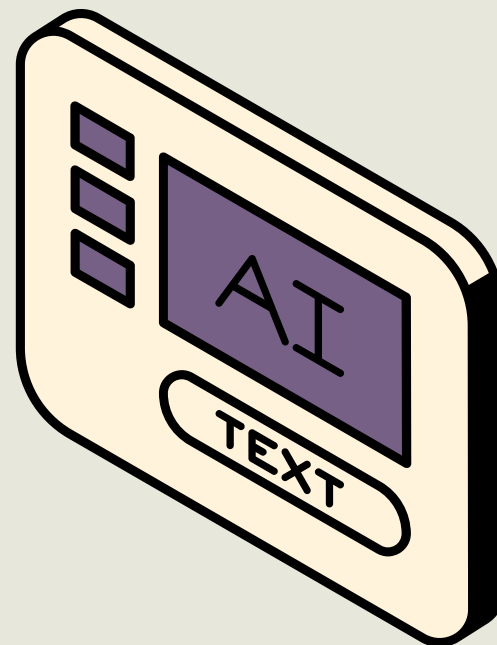
Design Systems: Storybook and Tailwind UI tokens that map design properties directly to code variables.

Dev Mode Inspection: Frontend engineers use Figma Dev Mode to inspect layout constraints, copy precise CSS/Tailwind classes, and export assets.

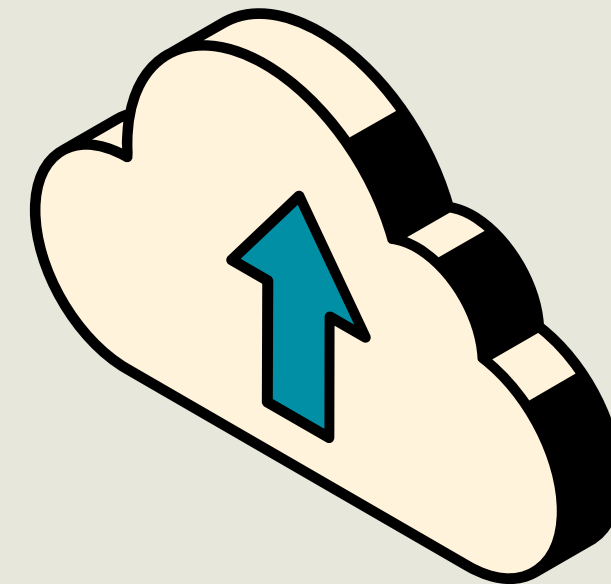


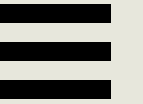
The Seamless Handoff Workflow

Dev Mode Inspection: Frontend engineers use Figma Dev Mode to inspect layout constraints, copy precise CSS/Tailwind classes, and export assets.



Component Implementation: Developers build reusable components matching the design system precisely, minimizing design discrepancies.

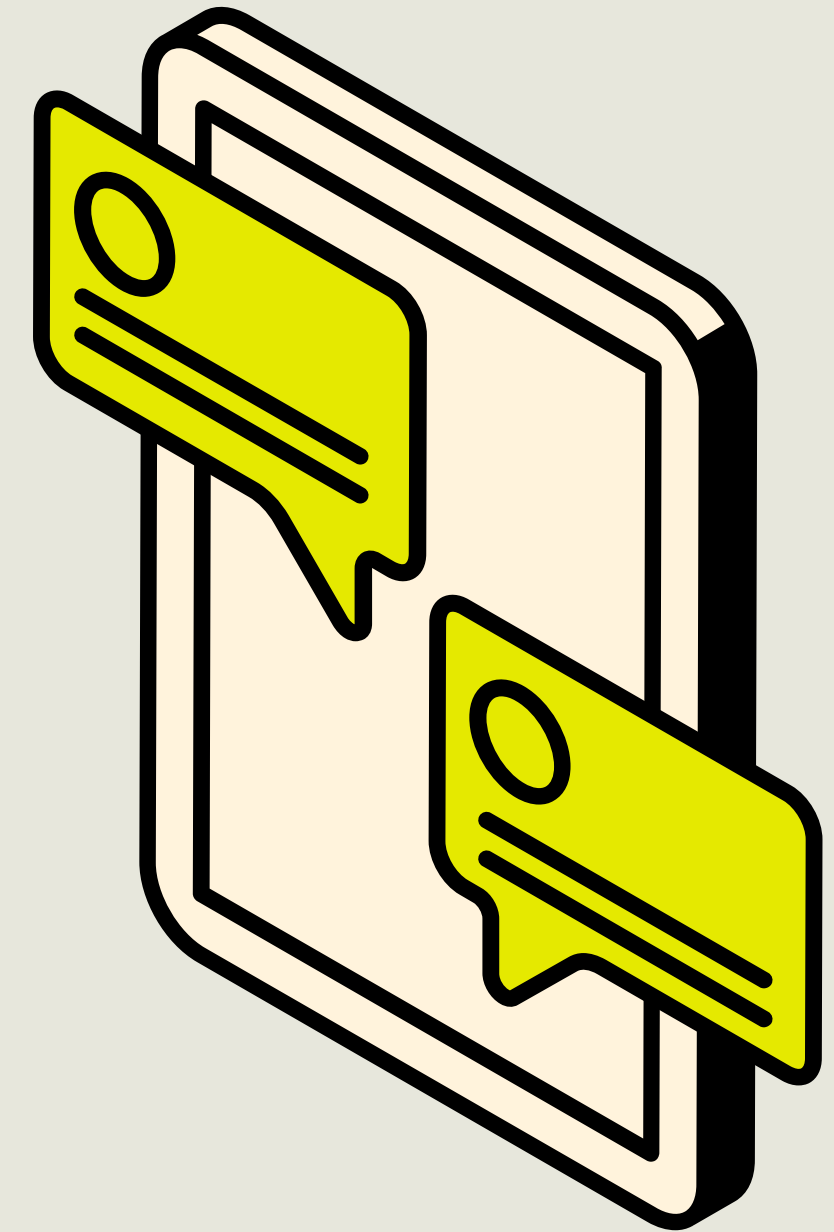


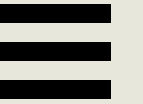


In today's world, technology is woven into nearly every aspect of life. From online shopping and mobile banking to smart homes and wearable health devices, innovations simplify daily routines and expand opportunities. Cars are becoming electric, offices rely on digital collaboration, and even leisure is shaped by streaming platforms and gaming. Technology no longer sits outside human life—it lives inside it, constantly shaping decisions and behaviors.

Technology in

Daily Life





Essential Developer Tools

IDEs & Editors

VS Code and JetBrains for efficient coding..

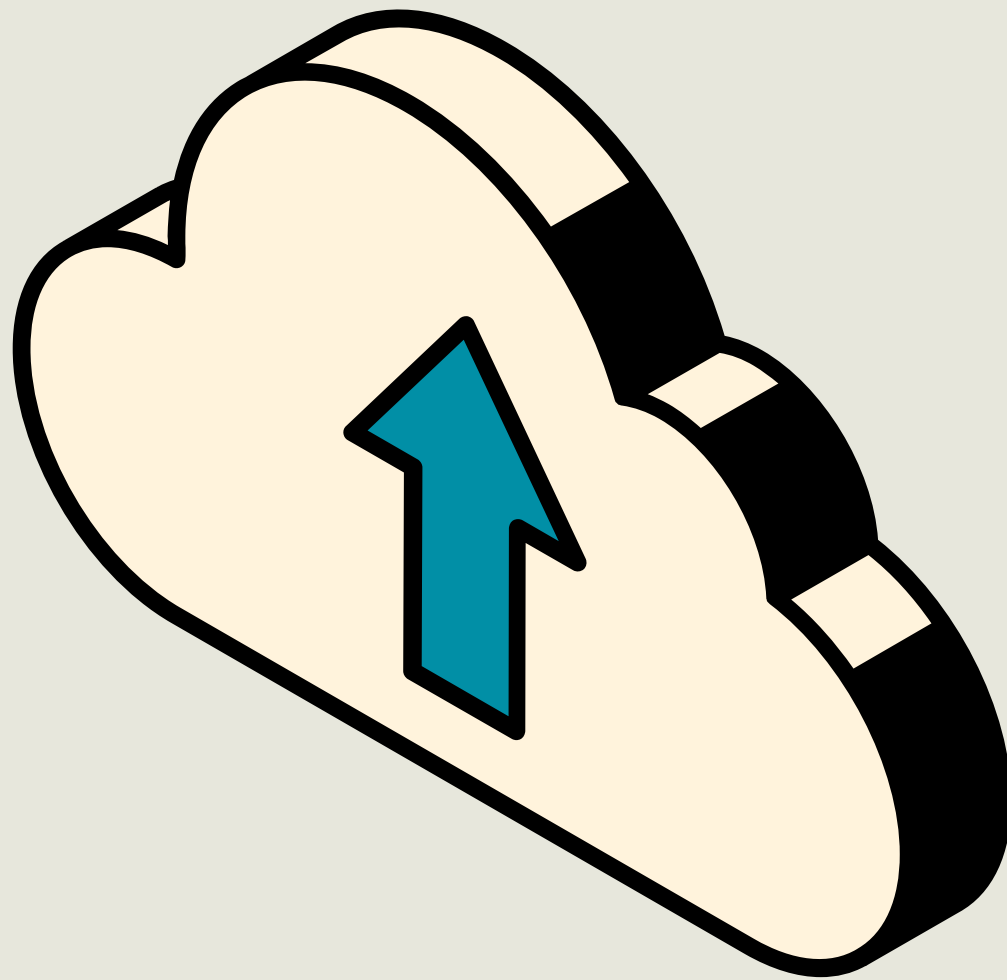
Version Control: Git with GitHub/GitLab for code management and collaboration.

API Testing: Postman, Insomnia, or Bruno for testing APIs.

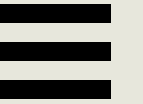
Project Management: Frontend engineers use Figma Dev Mode to inspect layout constraints, copy precise CSS/Tailwind classes, and export assets.



UI/UX Design and the Developer Handoff



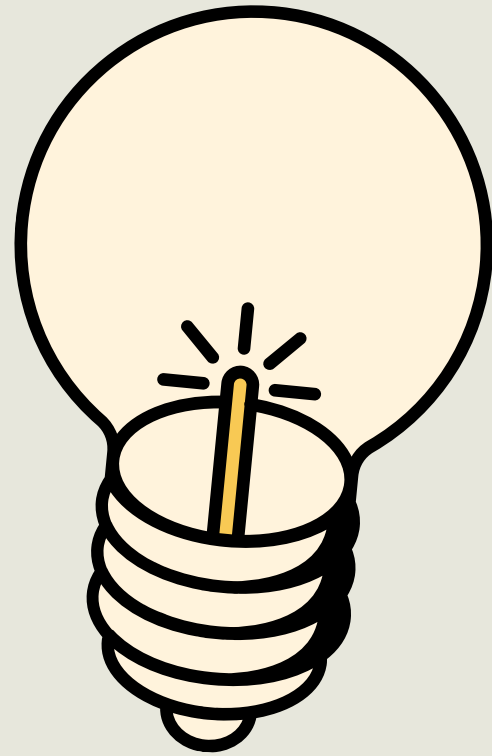
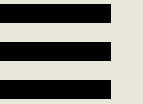
- **CI/CD:** GitHub Actions, GitLab CI, and CircleCI automate testing and deployment.
- **IaC:** Terraform manages cloud infrastructure through configuration files.
- **Containers & Serverless:** Docker, AWS Lambda, and Vercel enable flexible scaling.



The AI-Augmented Developer

- AI Assistants: Copilot and AI agents help generate, test, and explain code.
- Faster Prototyping: AI tools speed up MVP and UI development.
- Paradigm Shift: Developers focus more on problem-solving, system design, security, and code review.





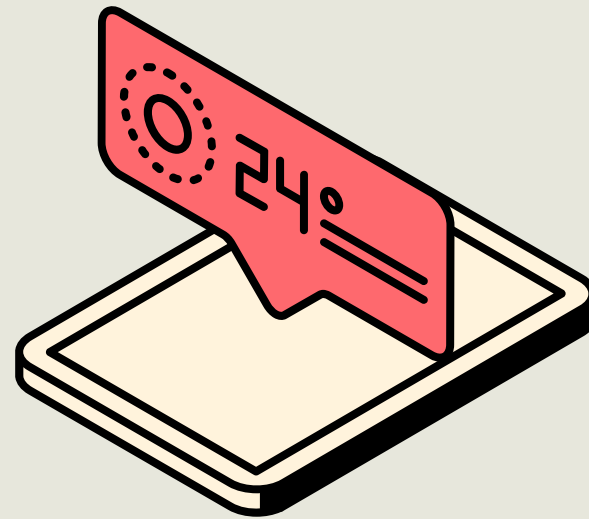
Conclusion & Best Practices

- Keep Learning: Master core principles while adapting to new tools.
- Collaborate: Connect PMs, designers, developers, and DevOps.
- Automate: Automate testing, deployment, and formatting to improve efficiency.



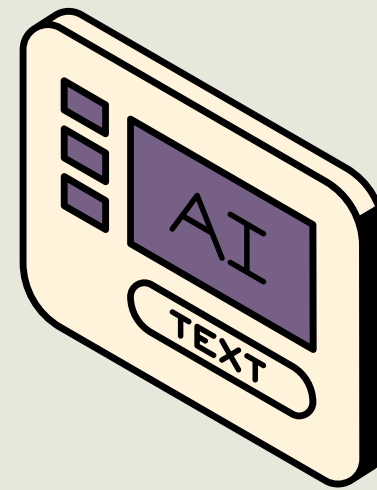


Project Economics & Stack Pricing



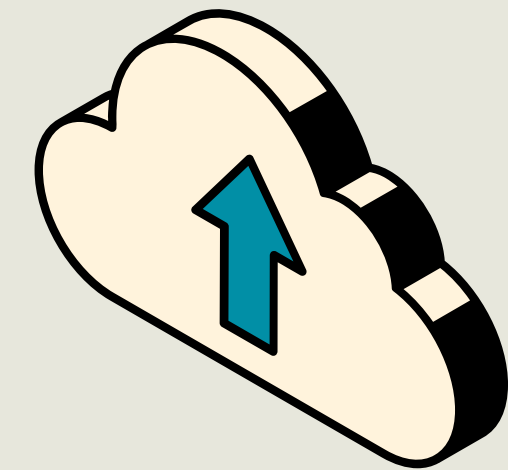
Stack & Infrastructure:

Open-source lowers costs; enterprise/cloud systems increase expenses.



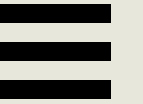
Team & Design:

Specialized roles and detailed UI/UX can raise costs and timelines.



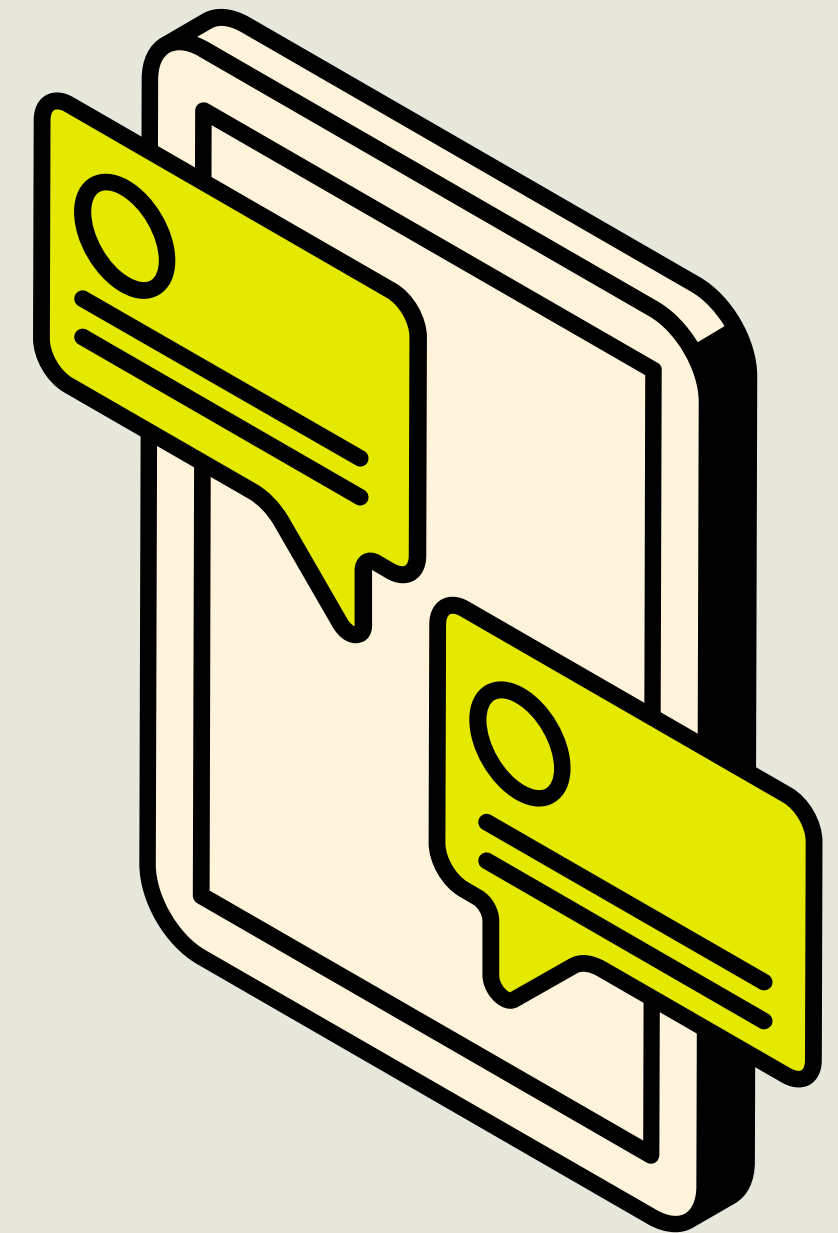
Long-Term Costs:

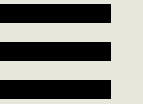
Poor stack choices create technical debt; security and compliance add expenses.



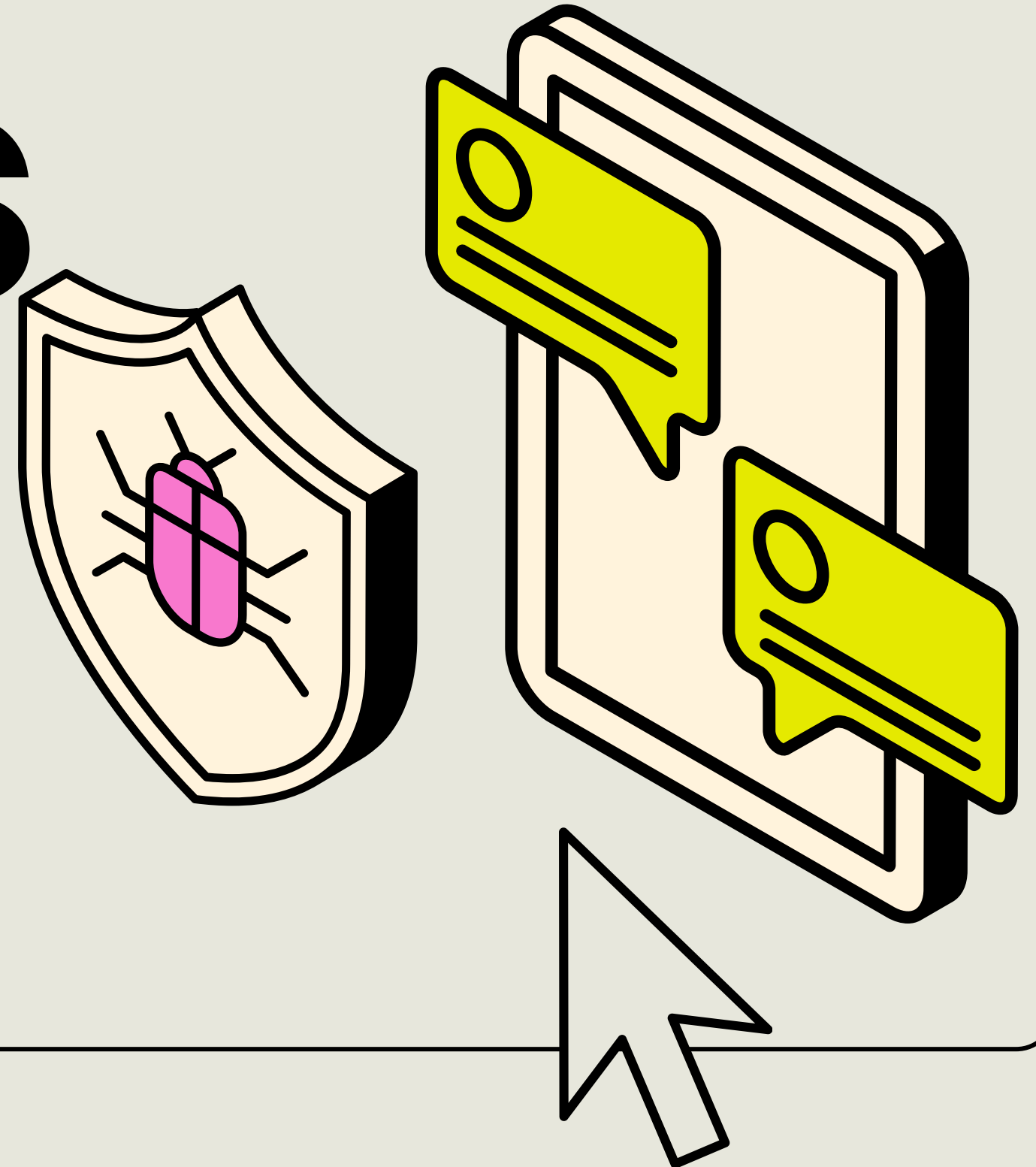
Key Takeaway:

Software budgeting scales directly with scalability, architecture, and security. Choosing a stack is a financial decision, not just a technical preference.





Questions & Answers



CERTIFICATE OF COMPLETION

PROUDLY PRESENTED TO



KHENARD VINCE ALBAYTAR

In appreciation for sharing valuable insights and active participation during the seminar entitled: **Modern Development: Programming stacks, Roles and Tools**. Held at OLFU Antipolo Campus SPCB 101 on the 12th of September for BSIT students, focusing on the evolving roles, tools, and technologies shaping today's software development landscape.

Benjamin B. Gandeza Jr. DIT(c)

CTIC411-Adviser

A stylized black ink signature of Edward N. Cruz, MIT.

Edward N. Cruz, MIT

CTIC411-Adviser



We would like to thank the following people to make this seminar happen

Khenard Vince Albaytar – Speaker

Driefen Alfonso – CERT Maker

Troy Almocera – Documentation

Dhane Lorenz Andrade – PPT

Jayro Apostol – Opening

Karl Niño Arcilla – Documentation

Gabrielle Atienza – Registration

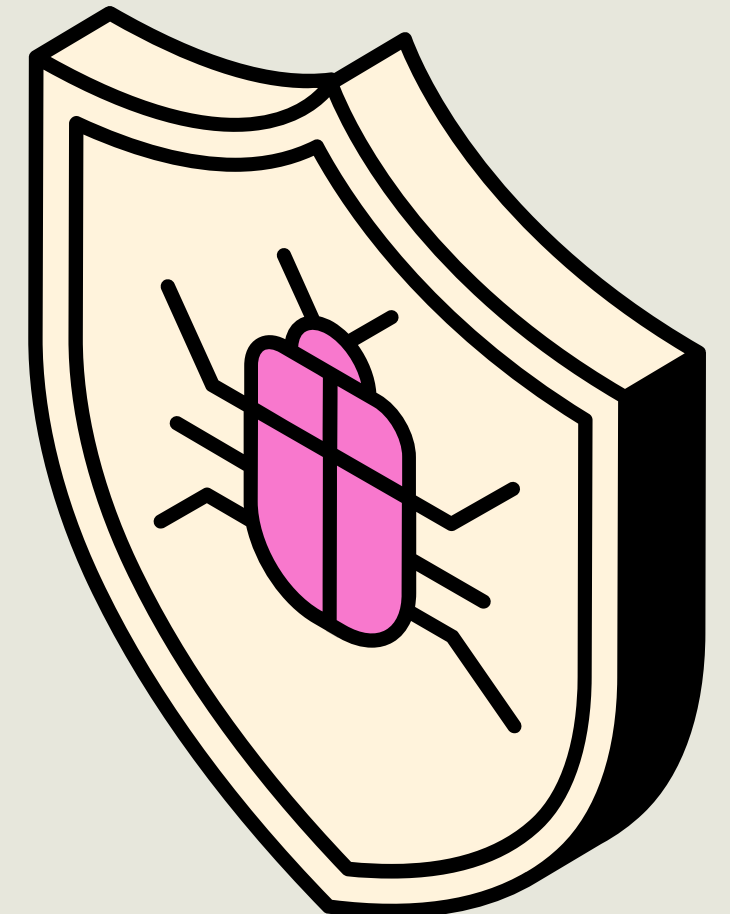
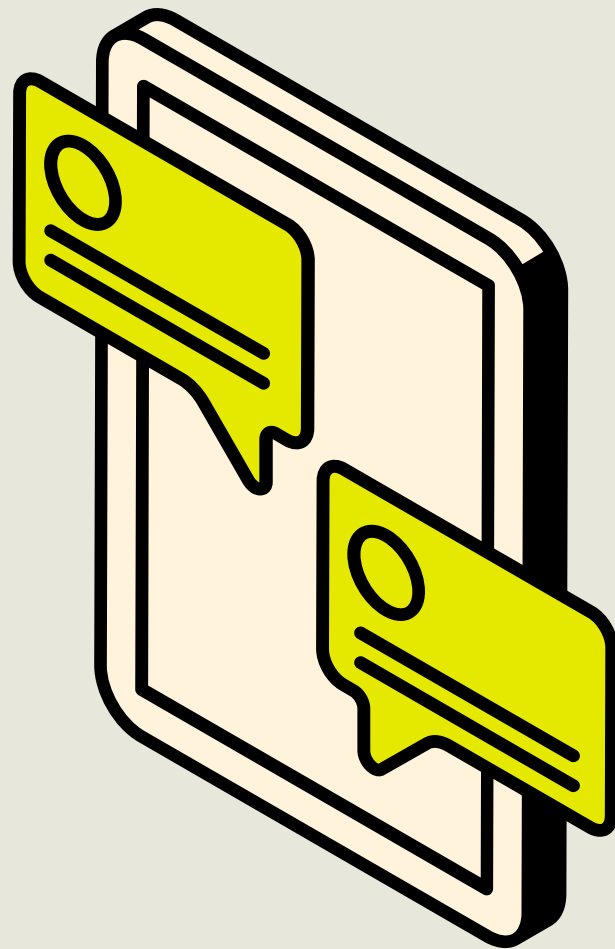
Vivienne Balasabas – Tech Support

Justine Cadillo – Closing

Vincent Capidos – Tech Support

Andrei Cariño – Host

Sprewell Conde – PPT Maker



EVALUATION

Thank You
EVERYONE!!!